

Complex Analysis

Our goal is to develop the surprisingly rich theory of complex valued functions of one complex variable $z \mapsto f(z)$ and the theory of integrating such functions along paths. Why should we care?

- (i) Polynomials $p(z) = a_0 + a_1z + \cdots + a_dz^d$ are of interest in geometry and number theory.
- (ii) Power series. For example, $s \mapsto \sum_{n \in \mathbb{N}} 1/n^s$ defines a complex differentiable function when $\operatorname{Re}(s) > 1$. This matters even if you primarily care about real values.
- (iii) There are connections to harmonic functions. These arise when describing soap bubbles or in partial differential equations.
- (iv) We can use complex methods to solve classical integrals or differential equations.

1 Complex differentiation

Some background language first. Let $z \in \mathbb{C}$. We write $x = \operatorname{Re}(z)$ and $y = \operatorname{Im}(z)$. The argument $\theta = \arg(z)$ is well-defined up to adding $2\pi\mathbb{Z}$. We usually use the principal argument $\operatorname{Arg}(z) \in (-\pi, \pi]$.

An open ball or open disc in $\mathbb{C}$ denoted $B(a, r) = D(a, r) = \{z \in \mathbb{C} : |z - a| < r\}$. We write $\mathbb{D} = B(0, 1)$. The closed ball is $\bar{B}(a, r) = \{z \in \mathbb{C} : |z - a| \leq r\}$. We write $\mathbb{C}^* = \mathbb{C} \setminus \{0\}$.

Recall that a set $U \subseteq \mathbb{C}$ is open if it contains an open disc about each of its points. That is, for all $w \in U$, there exists $\varepsilon > 0$ such that $B(w, \varepsilon) \subseteq U$.

Definition 1.1 – Path

A path in U is a continuous map $\gamma : [a, b] \rightarrow U$. We often use $[a, b] = [0, 1]$.

- (i) We say that γ is a closed path if $\gamma(a) = \gamma(b)$.
- (ii) We say that γ is C^1 if it is continuously differentiable. That is, for all $t \in [a, b]$, $\gamma'(t)$ exists and that function is continuous.
- (iii) We say that a path γ is simple if γ is injective except perhaps at its end points. That is, if $t_1 \neq t_2 \in [a, b]$ and $\gamma(t_1) = \gamma(t_2)$, then $\{t_1, t_2\} = \{a, b\}$.

Definition 1.2 – Path-connectedness and domains

We say that U is path-connected if for all $x, y \in U$, there exists a continuous $\gamma : [0, 1] \rightarrow U$ such that $\gamma(0) = x$ and $\gamma(1) = y$. A domain in $\mathbb{C}$ is a non-empty open path-connected subset of $\mathbb{C}$.

Definition 1.3 – Differentiable, holomorphic and entire

Let $U \subseteq \mathbb{C}$ be open.

- (i) We say that $f : U \rightarrow \mathbb{C}$ is differentiable at $w \in U$ if $f'(w) = \lim_{z \rightarrow w} \frac{f(z) - f(w)}{z - w}$ exists.
- (ii) We say that $f : U \rightarrow \mathbb{C}$ is holomorphic at $w \in U$ if there exists $\varepsilon > 0$ such that f is differentiable on $B(w, \varepsilon) \subseteq U$.
- (iii) If $U = \mathbb{C}$ and $f : \mathbb{C} \rightarrow \mathbb{C}$ is holomorphic everywhere, then f is said to be entire.

Some texts refer to holomorphic functions as analytic functions.

The usual rules for differentiating sums, products and inverses apply exactly as in real analysis, with exactly the same proofs.

Any $f : U \rightarrow \mathbb{C}$ can be written as $f(z) = f(x + iy) = u(x, y) + iv(x, y)$ where $u, v : U \rightarrow \mathbb{R}$ are the real and imaginary parts of f . Recall that $u : U \rightarrow \mathbb{R}$ is differentiable at $(c, d) \in U$ with derivative $[Du]_{(c,d)} = (\lambda, \mu)$ if and only if

$$\frac{u(x, y) - u(c, d) - [\lambda(x - c) + \mu(y - d)]}{\sqrt{(x - c)^2 + (y - d)^2}} \rightarrow 0$$

as $(x, y) \rightarrow (c, d)$.

Proposition 1.1 – Cauchy-Riemann equations

Let $f : U \rightarrow \mathbb{C}$ be defined on an open set U . Write $f = u + iv$. Then, f is differentiable at $w = c + id \in U$ with $f'(w) = p + iq$ if and only if u and v are both differentiable at (c, d) and

$$\frac{\partial u}{\partial x} = \frac{\partial v}{\partial y} = p \text{ and } -\frac{\partial u}{\partial y} = \frac{\partial v}{\partial x} = q.$$

Then, $f'(w) = u_x(c, d) + iv_x(c, d)$.

Proof. f is differentiable at w with derivative $p + iq$ if and only if

$$\lim_{z \rightarrow w} \frac{f(z) - f(w) - (z - w)(p + iq)}{z - w} = 0.$$

One can check that

$$\lim_{z \rightarrow a} \frac{f(z)}{g(z)} = 0 \iff \lim_{z \rightarrow a} \frac{f(z)}{|g(z)|} = 0.$$

Note that $(z - w)(p + iq) = p(x - c) - q(y - d) + i[q(x - c) + p(y - d)]$. By taking real and imaginary parts, the limit is zero if and only if

$$\lim_{(x,y) \rightarrow (c,d)} \frac{u(x, y) - u(c, d) - [p(x - c) - q(y - d)]}{\sqrt{(x - c)^2 + (y - d)^2}} = 0$$

and

$$\lim_{(x,y) \rightarrow (c,d)} \frac{v(x, y) - v(c, d) - [q(x - c) + p(y - d)]}{\sqrt{(x - c)^2 + (y - d)^2}} = 0.$$

This is exactly the condition for u to be differentiable with $[Du]_{(c,d)} = (p, -q)$ and v to be differentiable with $[Dv]_{(c,d)} = (q, p)$. $\square$

- (i) Write $f = u + iv$. If $u_x = v_y$ and $u_y = -v_x$ at a point w , we cannot conclude that f is differentiable at w . We need to know that u and v are differentiable.

- (ii) If the partial derivatives u_x, u_y, v_x, v_y exist and are continuous in an open neighbourhood of w , then the Cauchy-Riemann equations holding does imply complex differentiability at w .

Example 1.1

- (i) Polynomials are entire.
- (ii) If p and q are polynomials and $U \subseteq \mathbb{C} \setminus \{\text{zeroes of } q\}$, then the rational function $p(z)/q(z)$ is differentiable on U .
- (iii) $f(z) = |z|$ is not differentiable anywhere on $\mathbb{C}$. $f = u + iv$ with $u = \sqrt{x^2 + y^2}$ and $v = 0$. If $(x, y) \neq (0, 0)$, then

$$u_x = \frac{x}{\sqrt{x^2 + y^2}} \text{ and } u_y = \frac{y}{\sqrt{x^2 + y^2}} \text{ and } v_x = v_y = 0$$

so the Cauchy-Riemann equations fail. At $(0, 0)$, $[f(h) - f(0)]/h = |h|/h$ and the limit does not exist.

Later, we will see that if $f(z)$ is holomorphic on U , then $f'(z)$ is also holomorphic on U . Differentiating the Cauchy-Riemann equations, we have $u_{xx} = v_{yx} = -u_{yy}$ and $v_{xx} = -u_{yx} = -v_{yy}$ so u and v are harmonic functions. In fact, any harmonic function on an open set in $\mathbb{R}^2$ is locally $\text{Re}(f)$ for some holomorphic f .

2 Conformal maps

Proposition 2.1

Let $U \subseteq \mathbb{C}$ be a domain. Suppose $f : U \rightarrow \mathbb{C}$ is holomorphic and $f'(z) = 0$ on U . Then, f is constant.

Proof. We will use without proof the following elementary topological fact. If U is a domain and $\gamma : [0, 1] \rightarrow U$ is a path with $\gamma(0) = a$ and $\gamma(1) = b$, then there is another path $\tilde{\gamma} : [0, 1] \rightarrow U$ with $\tilde{\gamma}(0) = a$ and $\tilde{\gamma}(1) = b$ such that $\tilde{\gamma}$ is composed of finitely many segments, each parallel to the x -axis or y -axis. Note that $\tilde{\gamma}$ is piecewise C^1 . Given this, we know $u_x = v_y$ and $u_y = -v_x$ hold. In fact, $u_x = u_y = v_x = v_y = 0$ since $f'(z) = 0$ on U . The usual mean value theorem from real analysis shows that u and v are constant along the segments of $\tilde{\gamma}$ so $f(a) = f(b)$. $\square$

For all $t \in [0, 1]$, there exists $\varepsilon(t) > 0$ such that $B(\gamma(t), \varepsilon(t)) \subseteq U$ since U is open. Covering compactness implies that there exist finitely many $0 \leq t_1 < t_2 < \dots < t_n \leq 1$ such that $\text{image}(f) \subseteq \bigcup_{i=1}^n B(\gamma(t_i), \varepsilon(t_i))$. We can then check that $\tilde{\gamma}$ exists.

Definition 2.1 – Conformal map

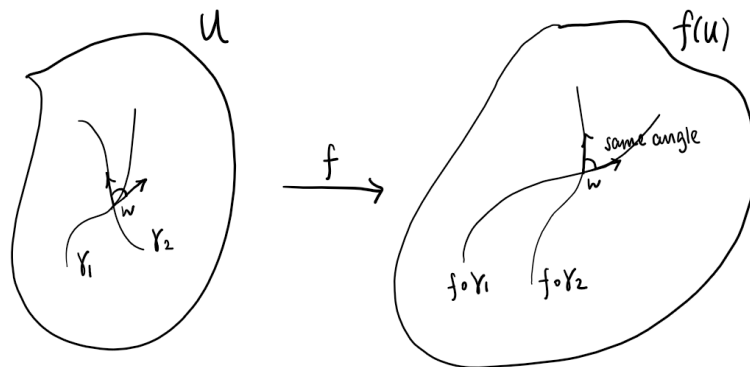
If f is holomorphic at a point w and $f'(w) \neq 0$, we say that f is conformal at w .

This is a geometric property of f . Suppose U is a domain and $f : U \rightarrow \mathbb{C}$ is conformal at $w \in U$. Take $\gamma_i : (-\varepsilon, \varepsilon) \rightarrow U$ with $\gamma_i(0) = w$ and $\gamma'_i(0) \neq 0$ so that γ_i have non-zero tangent vectors at w . Consider two different paths γ_1 and γ_2 . The angle between these vectors is $\arg(\gamma'_1(0)) - \arg(\gamma'_2(0)) = \arg(\gamma'_1(0)/\gamma'_2(0))$. Now consider applying f to these two paths.

$$\frac{(f \circ \gamma_1)'(0)}{(f \circ \gamma_2)'(0)} = \frac{f'(w) \gamma'_1(0)}{f'(w) \gamma'_2(0)} = \frac{\gamma'_1(0)}{\gamma'_2(0)}$$

and so the angle between $f \circ \gamma_1$ and $f \circ \gamma_2$ at $f(w)$ (that is, the angle between their tangent lines) is the same

as for γ_1 and γ_2 . Hence, f is conformal at w implies that f preserves angles in this sense.



Definition 2.2 – Conformal equivalence

If U and V are open in $\mathbb{C}$, and $f : U \rightarrow V$ is a holomorphic bijection which is conformal on all for U , we say that f is a conformal equivalence and that U and V are conformally equivalent.

In this case, the inverse function theorem says that f is locally invertible, and that the local inverse is complex differentiable. Since f is a bijection, it is globally invertible. The chain rule shows that f^{-1} is conformal.

Example 2.1

(i) A linear map $f(z) = az + b$ with $a \neq 0$ is a conformal equivalence $\mathbb{C} \rightarrow \mathbb{C}$.

(ii)

$$\begin{aligned} \{z \in \mathbb{C} \mid 0 < \arg(z) < \pi/n\} &\rightarrow \{z \in \mathbb{C} \mid \operatorname{Im}(z) > 0\} = \mathcal{H} \\ z &\mapsto z^n \end{aligned}$$

is conformal where $\mathcal{H}$ is the upper half-plane.

(iii)

$$z \in \mathcal{H} \iff z \text{ is closer to } i \text{ than to } -i \iff |z - i| < |z + i| \iff \left| \frac{z - i}{z + i} \right| < 1.$$

The Möbius map $z \mapsto (z - i)/(z + i)$ takes $\mathcal{H} \rightarrow \mathbb{D}$. Moreover, $f'(z) = 2i/(z + i)^2$ is non-zero for $z \in \mathcal{H}$ so f is a conformal equivalence $\mathcal{H} \rightarrow \mathbb{D}$.

Recall the Möbius map $A : z \mapsto (az + b)/(cz + d)$ with $ad - bc \neq 0$. A defines a conformal equivalence $\mathbb{C} \setminus \{-d/c\} \rightarrow \mathbb{C} \setminus \{a/c\}$. However, it is much better to think of it as a conformal equivalence of the extended complex plane $\mathbb{C}_\infty = \mathbb{C} \cup \{\infty\}$ and we work in a coordinate $w = 1/z$ near $z = \infty$. Also recall that Möbius maps are triply transitive,

$$z \mapsto \frac{z - z_1}{z - z_3} \frac{z_2 - z_3}{z_2 - z_1}$$

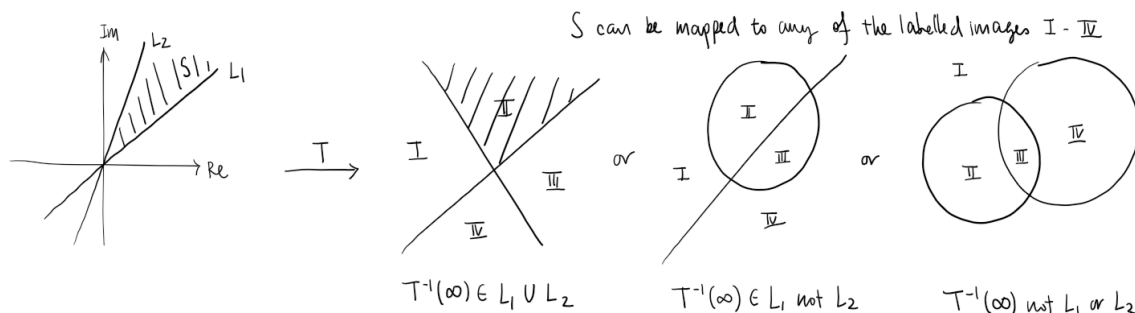
sends $(z_1, z_2, z_3) \mapsto (0, 1, \infty)$. Recall also that Möbius maps send circles and lines to circles and lines. We can think of lines as circles through ∞ . We can also think of $\mathbb{C}_\infty$ as a two-dimensional sphere via the stereographic projection

$$z \mapsto \left(\frac{z + \bar{z}}{1 + |z|^2}, \frac{-i(z - \bar{z})}{1 + |z|^2}, \frac{|z|^2 - 1}{1 + |z|^2} \right) \in \{(u, v, w) \in \mathbb{R}^3 \mid u^2 + v^2 + w^2 = 1\}.$$

Example 2.2

(i) The exponential map $z \mapsto e^z = e^x e^{iy}$ sends vertical lines to circles.

(ii) Consider the possible images of a sector $\{\alpha < \arg(z) < \beta\}$ under a Möbius map.



(iii) The Joukowski transformation is the map

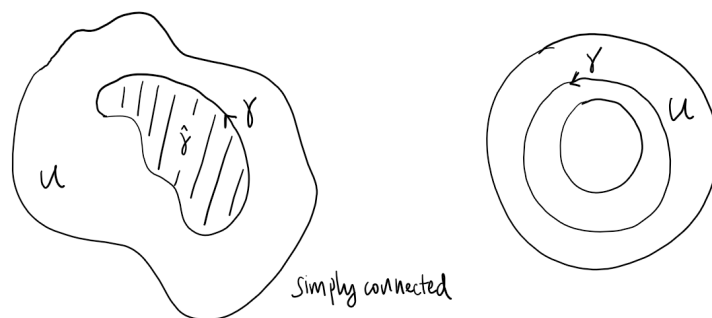
$$f : z \mapsto \frac{1}{2} \left(z + \frac{1}{z} \right) = \frac{z^2 + 1}{2z}.$$

Then $f'(z) = 1 - (z^2 + 1)/(2z^2)$ so f is holomorphic except at 0 and f is conformal except at $\{0, \pm 1\}$. The image of a circle passing through -1 and $-i$ resembles an aerofoil. The incompressible fluid equations are $\nabla \times \mathbf{v} = \mathbf{0}$ and $\mathbf{v} = \nabla \phi$ so ϕ is harmonic and ϕ is locally $\text{Re}(f)$ with f holomorphic. This is used to transfer questions about fluid flow from the right hand side to the left, which is simpler.

Let $S^1 = \partial \mathbb{D} = \{z \in \mathbb{C} \mid |z| = 1\}$.

Definition 2.3 – Simply connected domain

Let $U \subseteq \mathbb{C}$ be a domain. We say that U is simply connected if every continuous map $\gamma : S^1 \rightarrow U$ extends to a map $\hat{\gamma} : \mathbb{D} \rightarrow U$. That is, there exists a continuous $\hat{\gamma} : \mathbb{D} \rightarrow U$ such that $\hat{\gamma}|_{\partial \mathbb{D}} = \gamma$.



We have $U = \{z \mid r_1 < |z| < r_2\}$ and $\gamma(t) = re^{2\pi it}$ with $r \in (r_1, r_2)$ for the figure on the right. In this case, no $\hat{\gamma}$ exists so it is not simply connected.

Theorem 2.1 – Riemann mapping theorem

If $U \subsetneq \mathbb{C}$ is a proper subdomain and U is simply connected, then U is conformally equivalent to $\mathbb{D}$.

- (i) Being simply connected is invariant under conformal equivalence. Indeed, it is invariant under all homeomorphisms.
- (ii) In a few lectures time, we will see that $\mathbb{C}$ and $\mathbb{D}$ are not conformally equivalent.
- (iii) This theorem is very general. Even a region inside a fractal curve is conformally equivalent to $\mathbb{D}$.

3 Power series

A useful construction of holomorphic functions (and in some sense the universal construction as we will see later) is convergent power series. Recall that

- (i) a sequence f_n of functions on a set T converges uniformly to f on T if for all $\varepsilon > 0$, there exists $N > 0$ such that for all $n \geq N$ and $x \in T$, we have $|f_n(x) - f(x)| < \varepsilon$.
- (ii) the uniform limit of continuous functions is continuous.
- (iii) the Weierstrass M -test states that if $M_n \in \mathbb{R}_{>0}$ such that $0 \leq |f_n(x)| \leq M_n$ for all $x \in T$ and $\sum_{i=1}^{\infty} M_i < \infty$, then $\sum_{i=1}^N f_i(x) = F_N(x)$ converges uniformly as $N \rightarrow \infty$.
- (iv) if we let $(c_n) \subseteq \mathbb{C}$ be a sequence and $a \in \mathbb{C}$, then there is a unique radius of convergence $R \in [0, \infty]$ such that the series $z \mapsto \sum_{n \geq 0} c_n(z - a)^n$ converges absolutely in $|z - a| < R$ and diverges if $|z - a| > R$. If $0 < r < R$, then the series converges absolutely and uniformly on $\overline{B(a, r)}$.
- (v) the root test says that $R = 1/\lambda$ where $\lambda = \limsup_n |c_n|^{1/n}$. Equivalently, $R = \sup\{r \geq 0 \text{ such that } |c_n|r^n \rightarrow 0 \text{ as } n \rightarrow \infty\}$.

Theorem 3.1

Let $f(z) = \sum_{n=0}^{\infty} c_n(z - a)^n$ be a complex power series with radius of convergence $R > 0$. Then, f is holomorphic on $B(a, R)$ and $f'(z) = \sum_{n=1}^{\infty} n c_n(z - a)^{n-1}$. Moreover, f' also has radius of convergence R . Thus, f is infinitely differentiable on $B(a, R)$ and $f^{(n)}(a) = n! c_n$.

Proof. Without loss of generality, take $a = 0$. Since $n|c_n| \geq |c_n|$ for all $n \geq 1$, the radius of convergence of $\sum_{n=1}^{\infty} n c_n(z - a)^{n-1}$ is at most R . But if $0 < R_1 < R$ and $z \in B(0, R_1)$,

$$|n c_n z^{n-1}| = n |c_n| R_1^{n-1} \left| \frac{z}{R_1} \right|^{n-1} \quad n \left| \frac{z}{R_1} \right|^{n-1} \rightarrow 0 \text{ as } n \rightarrow \infty.$$

Comparing to $\sum_{n \geq 1} |c_n| R_1^{n-1}$, we see that our new series has radius of convergence $\geq R_1$. This is true for all $R_1 < R$ so the radius of convergence is R . Now suppose $|z|, |w| < \rho < R$. Consider $(f(z) - f(w))/(z - w)$ and its partial sum

$$\sum_{n=0}^N c_n \frac{z^n - w^n}{z - w} = \sum_{n=0}^N c_n \sum_{j=0}^{n-1} z^j w^{n-1-j}. \quad \left| c_n \sum_{j=0}^{n-1} z^j w^{n-1-j} \right| \leq |c_n| n \rho^{n-1}$$

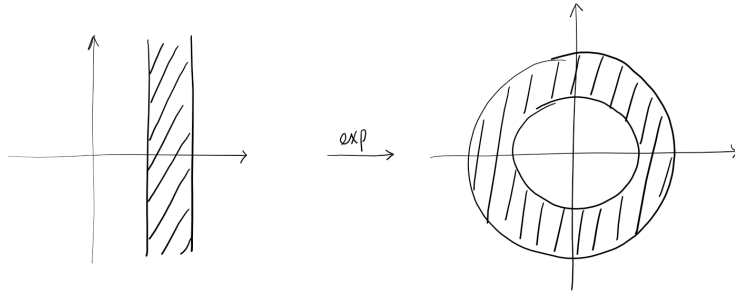
and the Weierstrass M -test says that the partial sum converges uniformly on $\{|z| < \rho, |w| < \rho\}$. Call the limit $g(z, w)$ and note that it is continuous. Then, $g(w, w) = \lim_{N \rightarrow \infty} \sum_{n=0}^N c_n n w^{n-1}$ so this does what we want. $\square$

Corollary 3.1

Let $\sum_{n=0}^{\infty} c_n(z - a)^n$ have radius of convergence $R > 0$. Suppose there exists $0 < \varepsilon < R$ such that $f \equiv 0$ on $B(a, \varepsilon) \subseteq B(a, R)$. Then, $f \equiv 0$ on $B(a, R)$.

Proof. We just saw that $c_n = f^{(n)}(a)/n!$ for $n \geq 1$. Hence, $f \equiv 0$. $\square$

Define $e^z = \exp(z) = \sum_{n=0}^{\infty} z^n/n!$ which has radius of convergence ∞ . By termwise differentiation, $d(e^z)/dz = e^z$. Since $e^z e^{-z} = 1$, $e^z \neq 0$ for all $z \in \mathbb{C}$.



The strip warps around the annulus infinitely often. So for $w \in \mathbb{C}^*$, there exists $z \in \mathbb{C}$ such that $e^z = w$, but there exists infinitely many such z .

We can also define the entire functions $\cos(z) = (e^{iz} + e^{-iz})/2$ and $\sin(z) = (e^{iz} - e^{-iz})/(2i)$.

Definition 3.1 – Branch of the logarithm

Let $U \subseteq \mathbb{C}^* = \mathbb{C} \setminus \{0\}$ be open. A continuous function $\lambda : U \rightarrow \mathbb{C}$ is a branch of the logarithm on U if $\exp(\lambda(z)) = z$ for all $z \in U$.

Example 3.1 – Principal branch of the logarithm

Let $U \subseteq \mathbb{C} \setminus \mathbb{R}_{\leq 0}$. Let $\text{Log} : U \rightarrow \mathbb{C}$ be defined by $\text{Log}(z) = \ln|z| + i\theta$ where $\theta = \text{Arg}(z) \in (-\pi, \pi)$. This is called the principal branch of the logarithm on U .

Proposition 3.1

Log is holomorphic on $\mathbb{C} \setminus \mathbb{R}_{\leq 0}$ with $(d/dz) \text{Log}(z) = 1/z$. If $|z| < 1$, then

$$\text{Log}(1+z) = \sum_{n=1}^{\infty} (-1)^n \frac{z^n}{n} = z - \frac{z^2}{2} + \frac{z^3}{3} - \dots$$

Proof. Log is continuous on $\mathbb{C} \setminus \mathbb{R}_{\leq 0}$ and $\exp$ is continuous on $\mathbb{C}$. If $z = e^x$ and $w = e^y$, then

$$\frac{\text{Log}(z) - \text{Log}(w)}{z - w} = \frac{x - y}{e^x - e^y} \rightarrow \frac{1}{e^y} = \frac{1}{w}$$

as $x \rightarrow y$. Hence, Log is differentiable with $(d/dz) \text{Log}(z) = 1/z$. So,

$$\begin{aligned} \frac{d}{dz} \text{Log}(1+z) &= \frac{1}{1+z} = 1 - z + z^2 - \dots && \text{converges for } |z| < 1 \\ \frac{d}{dz} \sum_{n=1}^{\infty} (-1)^n \frac{z^n}{n} &= \frac{1}{1+z} && \text{by termwise differentiation.} \end{aligned}$$

Hence, $\text{Log}(1+z) = \sum_{n=1}^{\infty} (-1)^n z^n / n + \text{constant}$ and we can evaluate at 0 to see that the constant is zero. $\square$

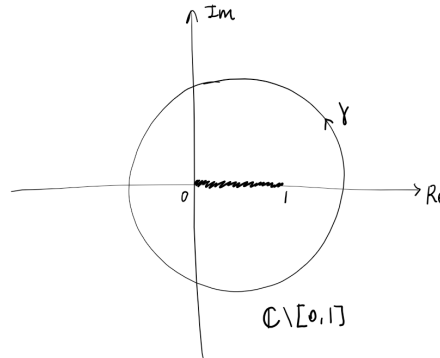
We will see later that there is no branch of the logarithm defined on $\mathbb{C}^*$. However, you can define a continuous branch of the logarithm on any simply connected domain not containing zero.

Definition 3.2 – Complex powers

For $\alpha \in \mathbb{C}$, the multivalued function z^α is by definition $\exp(\alpha \log(z))$ where $\log(z) = \ln|z| + i \arg(z)$.

If for $U \subseteq \mathbb{C}^*$, we have a branch of the logarithm specified on U , then using this we get a single-valued definition of z^α on U .

- (i) If $\alpha \in \mathbb{Z}$, it is single-valued by the periodicity of $\exp$.
- (ii) If $\alpha \in \mathbb{Q}$, it is finitely-many-valued. For example, $z^{1/2} = \exp((1/2) \log z)$ has two values at a general point, differing by -1 .
- (iii) We cannot define $\sqrt{z}$ globally on $\mathbb{C}$.
- (iv) The function $f(z) = z(z-1)$ admits a single-valued square root each of the domains $\mathbb{C} \setminus [0, 1]$ or $\mathbb{C} \setminus (\mathbb{R}_{\leq 0} \cup \mathbb{R}_{\geq 1})$. $(z(z-1))^{1/2} = \exp((1/2)(\log(z) + \log(z-1)))$. Along γ , the argument of $\log(z)$ and $\log(z-1)$ both jump by 2π . Since $\exp$ is 2π -periodic, we get a well-defined square root of $z(z-1)$.



Definition 3.3 – Branch point

A point $\rho \in \mathbb{C}$ is a branch point of a multivalued function ϕ if there is no continuous single-valued definition of ϕ in $B(\rho, \varepsilon)$ for any $\varepsilon > 0$.

Example 3.2

0 is a branch point of $\log$. $\{0, 1\}$ are branch points of $(z(z-1))^{1/2}$.

4 Contour integrals

We say that $f : [a, b] \rightarrow \mathbb{C}$ is Riemann integrable if both $\operatorname{Re}(f)$ and $\operatorname{Im}(f)$ are. Then,

$$\int_a^b f(t) dt = \int_a^b \operatorname{Re}(f(t)) dt + i \int_a^b \operatorname{Im}(f(t)) dt.$$

Note that

$$\left| \int_a^b f(t) dt \right| \leq \int_a^b |f(t)| dt.$$

For us, f will always be continuous and hence integrable.

Proposition 4.1

For $f : [a, b] \rightarrow \mathbb{C}$ continuous,

$$\left| \int_a^b f(t) dt \right| \leq \sup_{t \in [a, b]} |f(t)| (b - a).$$

Equality holds if and only if f is constant.

Proof. Let $\theta = \text{Arg}(\int_a^b f(t) dt)$ and $M = \sup_{t \in [a,b]} |f(t)|$. So,

$$\left| \int_a^b f(t) dt \right| = e^{-i\theta} \int_a^b f(t) dt = \int_a^b \text{Re}(e^{-i\theta} f(t)) dt \leq \int_a^b |e^{-i\theta} f(t)| dt \leq M(b-a).$$

The first inequality becomes an equality if $\text{Arg}(f) \equiv \theta$. The second inequality requires $|f| \equiv M$. Hence, equality holds if and only if f is constant. $\square$

Definition 4.1 – Contour

A contour is a simple closed path.

Definition 4.2 – Contour integrals

Suppose U is a domain and $f : U \rightarrow \mathbb{C}$ is continuous. Let $\gamma : [a, b] \rightarrow U$ be a C^1 -smooth curve.

$$\int_{\gamma} f(z) dz = \int_a^b f(\gamma(t)) \gamma'(t) dt$$

The definition extends in the obvious way to paths. We sometimes write $\int_{\gamma} f$ for $\int_{\gamma} f(z) dz$.

(i) $\int_{\gamma} (c_1 f_1 + c_2 f_2) = c_1 \int_{\gamma} f_1 + c_2 \int_{\gamma} f_2.$

(ii) If $a \leq \tilde{a} \leq b$, then $\int_{\gamma} f = \int_{\gamma|_{[a, \tilde{a}]}} f + \int_{\gamma|_{[\tilde{a}, b]}} f.$

(iii) Contour integrals are independent of parameterisation. Let $\phi : [a', b'] \rightarrow [a, b]$ be C^1 with $\phi(a') = a$ and $\phi(b') = b$. If $\delta = \gamma \circ \phi$, then $\int_{\delta} f = \int_{\gamma} f$. This is the usual chain rule.

(iv) If $(-\gamma)(t) = \gamma(-t)$ so $-\gamma : [-b, -a] \rightarrow U$, then $\int_{-\gamma} f = - \int_{\gamma} f.$

Set $\text{length}(\gamma) = \int_a^b |\gamma'(t)| dt$. This is not invariant under reparameterisation. It is if the reparameterisation is monotone.

Example 4.1

Let $U = \mathbb{C}^*$ and $f(z) = z^n$ with $n \in \mathbb{Z}$. Let $\phi : [0, 2\pi] \rightarrow U$ sending $t \mapsto e^{it}$. Then,

$$\int_{\gamma} f(z) dz = \begin{cases} 2\pi i & \text{if } n = -1 \\ 0 & \text{otherwise} \end{cases}$$

since

$$\int_{\gamma} f(z) dz = \int_0^{2\pi} e^{nit} (ie^{it}) dt = i \int_0^{2\pi} e^{i(n+1)t} dt.$$

Theorem 4.1 – Fundamental theorem of calculus

Let U be a domain and $F : U \rightarrow \mathbb{C}$ be holomorphic with $F'(z)$ continuous on U . For a path $\gamma : [a, b] \rightarrow U$,

$$\int_{\gamma} F'(z) dz = F(\gamma(b)) - F(\gamma(a)).$$

In particular, if γ is closed, then the integral is zero.

Proof.

$$\int_{\gamma} F'(z) dz = \int_a^b F'(\gamma(t)) \gamma'(t) dt = \int_a^b (F \circ \gamma)'(t) dt = (F \circ \gamma)(b) - (F \circ \gamma)(a)$$

□

We will show later that the hypothesis $F'(z)$ is continuous on U always holds. For example, we already know that if F was defined by a power series, this is true.

Example 4.2

If $n \neq -1$, then $z^n = (d/dz)(z^{n+1}/(n+1))$ so $\int_{|z|=1} z^n dz = 0$. If $n = -1$, then $1/z = (d/dz)(\log z)$ where $\log$ was defined on a slit plane.

Definition 4.3 – Antiderivative

Suppose U is a domain and $f : U \rightarrow \mathbb{C}$ is continuous. If $F : U \rightarrow \mathbb{C}$ is holomorphic with $F'(z) = f(z)$, then we say that F is an antiderivative for f on U .

Example 4.3

$f(z) = 1/z$ has no antiderivative on $\mathbb{C}^*$.

5 Cauchy's theorem

Lemma 5.1

If γ is a C^1 -path and f is continuous, then

$$\left| \int_{\gamma} f(z) dz \right| \leq \text{length}(\gamma) \sup_{\gamma} |f|.$$

Proof.

$$\left| \int_{\gamma} f(z) dz \right| = \left| \int_a^b f(\gamma(t)) \gamma'(t) dt \right| \leq \int_a^b |f(\gamma(t))| |\gamma'(t)| dt \leq \sup_{t \in [a,b]} |f(\gamma(t))| \text{length}(\gamma)$$

□

Lemma 5.2 – Converse to the Fundamental Theorem of Calculus

Let $U \subseteq \mathbb{C}$ be a domain and $f : U \rightarrow \mathbb{C}$ be continuous. Assume $\int_{\gamma} f(z) dz = 0$ for all piecewise C^1 closed curves γ in U . Then, $f(z)$ has an antiderivative $F(z)$ on U .

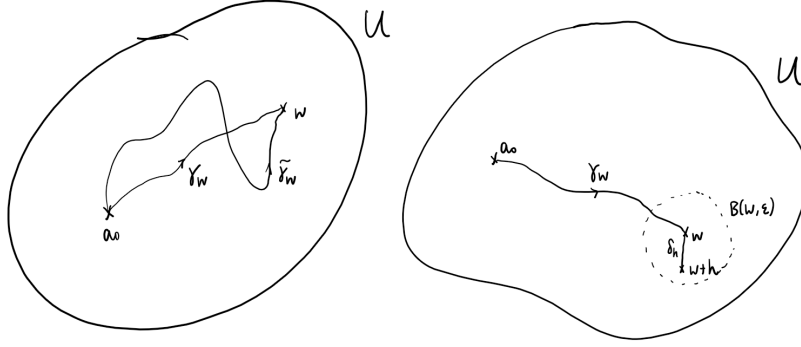
Proof. Pick $a_0 \in U$. Since U is a domain, it is path-connected. For $w \in U$, pick a piecewise C^1 -smooth γ_w from a_0 to w . Set $F(w) = \int_{\gamma_w} f(z) dz$. Our hypothesis that $\int_{\text{closed } \gamma} f(z) dz = 0$ implies that $F(w)$ does not depend on the choice of γ_w . This is because $\gamma_w \cup (-\tilde{\gamma}_w)$ is a closed curve. U is open so there exists $\varepsilon > 0$ such that $B(w, \varepsilon) \subseteq U$. If $|h| < \varepsilon$, let δ_h be the radial path from w to $w + h$. Then, $F(w + h) = \int_{\gamma_w * \delta_h} f(z) dz$ with $\gamma_w * \delta_h$ being the concatenated path. So,

$$F(w + h) = \int_{\gamma_w} f(z) dz + \int_{\delta_h} f(z) dz = F(w) + \underbrace{\int_{\delta_h} f(z) dz}_{hf(w)} + \int_{\delta_h} (f(z) - f(w)) dz.$$

Then,

$$\begin{aligned} \left| \frac{F(w+h) - F(w)}{h} - f(w) \right| &= \left| \frac{1}{h} \int_{\delta_h} (f(z) - f(w)) dz \right| \\ &\leq \frac{1}{|h|} \text{length}(\delta_h) \sup_{z \in \delta_h} |f(z) - f(w)| \\ &= \sup_{z \in \delta_h} |f(z) - f(w)|. \end{aligned}$$

As $h \rightarrow 0$, the right hand side tends to zero by the continuity of f . Hence, f is complex differentiable at w with $F'(w) = f(w)$. This is true for all $w \in U$.



□

Definition 5.1 – Convex and starlike domains

A domain U is convex if for all $p, q \in U$, the straight line segment $[p, q] \subseteq U$.

A domain U is starlike (with respect to $p_0 \in U$) if for all $q \in U$, the straight line segment $[p_0, q] \subseteq U$.

Note that a disc $\implies$ convex $\implies$ starlike $\implies$ path-connected.

Lemma 5.3

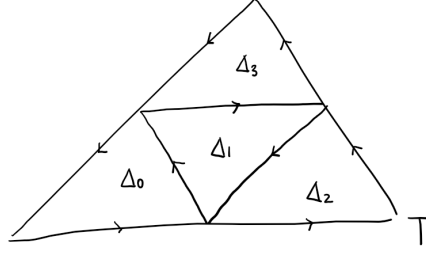
Let U be starlike and $f : U \rightarrow \mathbb{C}$ be continuous. Suppose $\int_{\gamma} f = 0$ whenever γ is a triangle in U . That is, γ is a piecewise C^1 path made of three edges of a Euclidean triangle, and the whole triangle including its interior lies in U . Then, f has an antiderivative on U .

Proof. Suppose U is starlike with respect to p_0 . For $w \in U$, let $\gamma_w = [p_0, w] \subseteq U$. Using the notation in the proof of Lemma 5.2, the paths $\gamma_w, \delta_h, \gamma_{w+h}$ form a triangle in U . We can then proceed as before. □

Theorem 5.1 – Cauchy's theorem for triangles

Let U be a domain and $T \subseteq U$ be a triangle in U . If $f : U \rightarrow \mathbb{C}$ is holomorphic, then $\int_{\partial T} f = 0$ where ∂T is the piecewise C^1 closed path given by the boundary of T .

Proof. Let $L = \text{length}(\partial T)$ and $I = |\int_{\partial T} f|$. By bisecting edges, we cut T into four smaller triangles. Orienting all four Δ_i anticlockwise, note that $\partial T = \partial \Delta_0 + \partial \Delta_1 + \partial \Delta_2 + \partial \Delta_3$. So, there exists i such that $|\int_{\partial \Delta_i} f| \geq I/4$. Label such Δ_i by $T_1 \subseteq T$. Iterate and inductively construct a nested sequence $T \supseteq T_1 \supseteq T_2 \supseteq \dots$ such that $|\int_{\partial T_n} f| \geq I/4^n$ and $\text{length}(\partial T_n) = 2^{-n}L$.



We claim that the intersection $\bigcap_{n \geq 1} T_n = \{w\}$ comprises of a single point w . Note that $\text{length}(\partial T_n) \rightarrow 0$ shows that $\text{diameter}(T_n) \rightarrow 0$, so there can be at most one point in this common intersection. That the intersection is not empty is either a basic property of sequential compactness or follows from Bolzano-Weierstrass applied to each coordinate (x_i, y_i) of a sequence of points of T_i .

Let $g(z) = (f(z) - f(w))/(z - w) - f'(w)$ which is continuous on U and vanishes at w .

$$\begin{aligned}
 \frac{I}{4^n} &\leq \left| \int_{\partial T_n} f \right| \\
 &= \left| \int_{\partial T_n} f(z) + \underbrace{f(w) - (z - w)f'(w)}_{\text{has an antiderivative so its } \int = 0 \text{ by FTC}} dz \right| \\
 &\leq \frac{L}{2^n} \sup_{z \in \partial T_n} |(z - w)g(z)| \\
 &\leq \frac{L^2}{4^n} \sup_{z \in \partial T_n} |g(z)| \text{ since } |z - w| \leq \text{diameter}(T_n) \leq \text{length}(\partial T_n) \\
 I &\leq L^2 \sup_{z \in \partial T_n} |g(z)|
 \end{aligned}$$

This tends to zero as $n \rightarrow \infty$ since $g(w) = 0$. □

Corollary 5.1 – Convex Cauchy

Let f be holomorphic on a starlike or convex domain U . Then, $\int_{\gamma} f = 0$ for any piecewise C^1 closed curve γ .

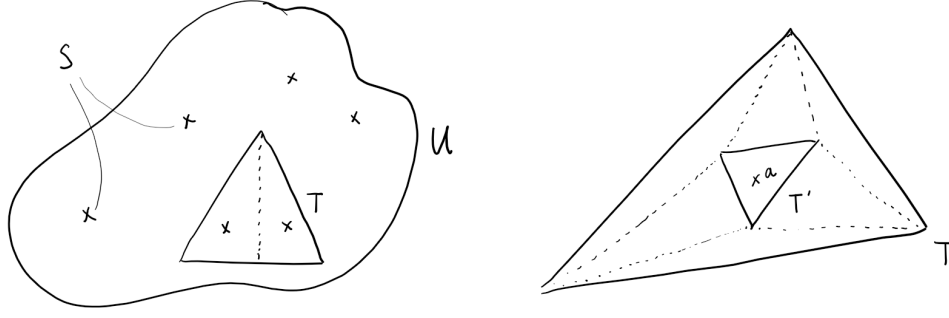
Proof. The argument for Theorem 5.1 implies that $\int_{\partial T} f = 0$ when T is a triangle in our domain U . Thus, f has an antiderivative F on U by Lemma 5.3. That is, F is holomorphic with $F'(z) = f(z)$ for all $z \in U$. Hence, $\int_{\gamma} f = \int_{\gamma} F' = 0$ by the Fundamental Theorem of Calculus since γ is closed. □

6 The Cauchy integral formula

Proposition 6.1

Let U be a domain and $f : U \rightarrow \mathbb{C}$ be continuous. Suppose there exists a finite set $S \subseteq U$ such that f is holomorphic on $U \setminus S$. If $T \subseteq U$ is a triangle, then $\int_{\partial T} f = 0$.

Proof.



Without loss of generality, suppose $S = \{\text{single point}\}$, by subdividing T if necessary. If T contains no points of S , use Theorem 5.1. And if $|S \cap T| > 1$, divide and add. Now suppose $a \in S$ and $a \in T' \subseteq T$. We can subdivide T into several smaller triangles, one of which is T' . Orienting all these triangles anticlockwise, we have $\int_{\partial T} f = \pm \int_{\partial T'} f$ by Theorem 5.1. Now,

$$\left| \int_{\partial T'} f \right| \leq \text{length}(\partial T') \sup_{z \in \partial T'} |f(z)| \leq \text{length}(\partial T') \sup_{z \in T} |f(z)|.$$

Note that $\sup_{z \in T} |f(z)|$ is constant since f is continuous on U and so is bounded on T . We can take T' to be as small as we like, so we can choose a sequence of inscribed triangles such that $\text{length}(\partial T') \rightarrow 0$. $\square$

Corollary 6.1

Suppose U is starlike and $S \subseteq U$ is finite. If $f : U \rightarrow \mathbb{C}$ is continuous and holomorphic on $U \setminus S$, then $\int_{\gamma} f = 0$ for all closed paths $\gamma \subseteq U$.

Proof. Exactly as deducing Corollary 5.1 from Theorem 5.1. $\square$

Theorem 6.1 – Cauchy integral formula

Let $U \subseteq \mathbb{C}$ be a domain and $f : U \rightarrow \mathbb{C}$ be holomorphic on U . Let $\overline{B(a, r)} \subseteq U$. For $z \in B(a, r)$ and $0 < \rho \leq r$ with $|z - a| < \rho$, we have

$$f(z) = \frac{1}{2\pi i} \int_{|w-a|=\rho} \frac{f(w)}{w-z} dw.$$

Proof. Let

$$g(w) = \begin{cases} (f(w) - f(z))/(w-z) & \text{if } w \neq z \\ f'(z) & \text{if } w = z. \end{cases}$$

This is continuous on $\overline{B(a, r)}$ and is holomorphic except perhaps at z . By Proposition 6.1,

$$\int_{\partial B(a, \rho)} g(w) dw = 0 \implies \int_{|w-a|=\rho} \frac{f(w)}{w-z} dw = \int_{|w-a|=\rho} \frac{f(z)}{w-z} dw.$$

Note that

$$\frac{1}{w-z} = \frac{1}{w-a} \frac{1}{1 - \frac{z-a}{w-a}} = \frac{1}{w-a} \sum_{n=0}^{\infty} \left(\frac{z-a}{w-a} \right)^n.$$

This series converges uniformly on $|w - a| = \rho$ since $|(z - a)/(w - a)| < 1$. So,

$$\int_{|w-a|=\rho} \frac{f(w)}{w-z} dw = f(z) \int_{|w-a|=\rho} \sum_{n=0}^{\infty} \frac{(z-a)^n}{(w-a)^{n+1}} dw = f(z) \sum_{n=0}^{\infty} \int_{|w-a|=\rho} \frac{(z-a)^n}{(w-a)^{n+1}} dw$$

where we used uniform convergence to exchange $\sum$ and $\int$. Recall that $\int_{|z|=1} z^n dz$ gave us something non-trivial only when $n = -1$. Changing values $w - a \mapsto w$, we see only the term with $n + 1 = 1$ contributes and gives $2\pi i$. Hence,

$$\int_{|w-a|=\rho} \frac{f(w)}{w-z} dz = 2\pi i f(z)$$

as required. $\square$

Corollary 6.2 – Mean value property

If $f : U \rightarrow \mathbb{C}$ is holomorphic on a domain U and $\overline{B(a, r)} \subseteq U$, then

$$f(a) = \int_0^1 f(a + re^{2\pi it}) dt.$$

That is, the value of f at a is the average of its values over a small linking circle.

Proof.

$$f(a) = \frac{1}{2\pi i} \int_{|w-a|=r} \frac{f(w)}{w-a} dw$$

by Theorem 6.1. Parameterise our contour of integration γ via $t \mapsto a + re^{2\pi it}$. Thus,

$$f(a) = \frac{1}{2\pi i} \int_{\gamma} \frac{f(\gamma(t))}{\gamma(t) - a} \gamma'(t) dt = \frac{1}{2\pi i} \int_{\gamma} \frac{f(\gamma(t))}{re^{2\pi it}} 2\pi i re^{2\pi it} dt$$

and the result follows. $\square$

Corollary 6.3 – Local maximum principle

If $f : B(a, r) \rightarrow \mathbb{C}$ is holomorphic and $|f(z)| \leq |f(a)|$ for all $z \in B(a, r)$, then f is constant.

Proof. For $0 < \rho < r$,

$$|f(a)| = \left| \int_0^1 f(a + \rho e^{2\pi it}) dt \right| \leq \sup_{|z-a|=\rho} |f(z)| \leq |f(a)|.$$

So, the inequalities must be equalities. That is, $|f(z)| = |f(a)|$ for all z with $|z - a| = \rho$. Since $\rho \in (0, r)$ is arbitrary, $|f|$ is constant. By Example sheet 1 Question 2, f is constant. $\square$

Corollary 6.4 – Liouville's theorem

Let $f : \mathbb{C} \rightarrow \mathbb{C}$ be entire (that is, holomorphic on $\mathbb{C}$). If f is bounded (that is, there exists M such that $|f(z)| \leq M$ for all $z \in \mathbb{C}$), then f is constant.

Proof. Let $z_1, z_2 \in \mathbb{C}$ and pick $R > 2 \max\{|z_1|, |z_2|\}$.

$$\begin{aligned} f(z_1) - f(z_2) &= \frac{1}{2\pi i} \int_{\partial B(0,R)} \left(\frac{f(w)}{w - z_1} - \frac{f(w)}{w - z_2} \right) dw \\ |f(z_1) - f(z_2)| &= \frac{1}{2\pi} \left| \int_{\partial B(0,R)} \frac{f(w)(z_1 - z_2)}{(w - z_1)(w - z_2)} dw \right| \\ &\leq \frac{1}{2\pi} (2\pi R) \frac{M|z_1 - z_2|}{(R/2)(R/2)} \rightarrow 0 \text{ as } R \rightarrow \infty \end{aligned}$$

Since z_1 and z_2 were arbitrary, f is constant. □

Corollary 6.5 – Fundamental theorem of algebra

Every complex polynomial of degree > 0 has a root in $\mathbb{C}$. So, it factors as a product of linear factors.

Proof. Let $p(z) = a_0 + a_1z + \cdots + a_dz^d$ where $a_d \neq 0$ and $d > 0$. Note that $|p(z)/z^d| \rightarrow |a_d|$ as $z \rightarrow \infty$, so $|p(z)| \rightarrow \infty$ as $z \rightarrow \infty$. Let $f(z) = 1/p(z)$. Suppose p never vanishes, then f is entire. Note that $|f(z)| \rightarrow 0$ as $z \rightarrow \infty$, so there exists $R > 0$ such that $|f(z)| \leq 1$ for all $|z| > R$. But, being continuous, f is bounded on $\overline{B(0, R)}$. Thus, f is bounded on $\mathbb{C}$, so f is constant by Liouville's theorem. Hence, p is constant, which is a contradiction. Therefore, p must vanish somewhere. □

7 Taylor and Morera

Theorem 7.1 – Taylor's theorem

If $f : B(a, r) \rightarrow \mathbb{C}$ is holomorphic, then f has a convergent power series representation

$$f(z) = \sum_{n=0}^{\infty} c_n (z - a)^n \quad \text{where } c_n = \frac{f^{(n)}(a)}{n!} = \frac{1}{2\pi i} \int_{\partial B(a, \rho)} \frac{f(w)}{(w - a)^{n+1}} dw$$

for any $0 < \rho < r$.

Proof. For $0 < \rho < r$ and $z \in \mathbb{C}$ is such that $|z - a| < \rho$, the Cauchy integral formula gives

$$\begin{aligned} f(z) &= \frac{1}{2\pi i} \int_{\partial B(a, \rho)} \frac{f(w)}{w - z} dw \\ &= \frac{1}{2\pi i} \int_{\partial B(a, \rho)} f(w) \frac{1}{(w - a) \left(1 - \frac{z - a}{w - a}\right)} dw \\ &= \frac{1}{2\pi i} \int_{\partial B(a, \rho)} f(w) \sum_{n=0}^{\infty} \frac{(z - a)^n}{(w - a)^{n+1}} dw \\ &= \sum_{n=0}^{\infty} \left(\frac{1}{2\pi i} \int_{\partial B(a, \rho)} \frac{f(w)}{(w - a)^{n+1}} dw \right) (z - a)^n. \end{aligned}$$

□

If $f : U \rightarrow \mathbb{C}$ is holomorphic on a domain U , then at $a \in U$, there exists $r > 0$ such that $B(a, r) \subseteq U$. Taylor's theorem gives us a power series representation on the ball, but not necessarily on all of U .

Corollary 7.1

If f is holomorphic on a domain U , then f is infinitely differentiable on U .

Proof. Combine Taylor's theorem with Theorem 3.1. □

Theorem 7.2 – Morera's theorem

Let U be a domain and $f : U \rightarrow \mathbb{C}$ be continuous. If $\int_{\gamma} f = 0$ for all piecewise- C^1 closed paths in U , then f is holomorphic on U .

Proof. By Lemma 5.2, $f = F'$ for some F holomorphic on U . By Corollary 7.1, f is holomorphic as well. □

Corollary 7.2

Let $f_n : U \rightarrow \mathbb{C}$ be a sequence of holomorphic functions on a domain U , and suppose $f_n \rightarrow f$ uniformly on U . Then, f is holomorphic and $f'(z) = \lim_{n \rightarrow \infty} f'_n(z)$ for $z \in U$.

Proof. Since we are only interested in the local limit of the derivatives, we can take $U = B(z, r)$ without loss of generality. If γ is a closed curve in $B(z, r)$, then $\int_{\gamma} f_n = 0$, and $\int_{\gamma} f_n \rightarrow \int_{\gamma} f$ by uniform convergence. Hence, $\int_{\gamma} f = 0$. Since f is continuous, f is holomorphic by Morera's theorem. For $0 < \rho < r$, by Taylor's theorem,

$$f'(z) = \frac{1}{2\pi i} \int_{\partial B(z, \rho)} \frac{f(w)}{(w-z)^2} dw$$

and similarly for each $f'_n(z)$. Thus,

$$\begin{aligned} |f'(z) - f'_n(z)| &= \frac{1}{2\pi} \left| \int_{\partial B(z, \rho)} \frac{f(w) - f_n(w)}{(w-z)^2} dz \right| \\ &\leq \frac{1}{2\pi} (2\pi\rho) \frac{1}{\rho^2} \sup_{|w-z|=\rho} |f(w) - f_n(w)| \rightarrow 0 \end{aligned}$$

as $n \rightarrow \infty$ by uniform convergence. □

Corollary 7.3

Suppose $f : U \rightarrow \mathbb{C}$ is continuous on a domain U , and for some finite set $S \in U$, f is holomorphic on $U \setminus S$. Then, f is holomorphic on U .

Proof. If $a \in S$, there exists $\varepsilon > 0$ such that $B(a, \varepsilon) \subseteq U$ and $B(a, \varepsilon) \cap S = \{a\}$, since S is finite. By Corollary 6.1, $\int_{\gamma} f = 0$ for all closed piecewise- C^1 paths γ inside $B(a, \varepsilon)$. Since f is continuous, f is holomorphic on $B(a, \varepsilon)$ by Morera's theorem. Hence, we are done. □

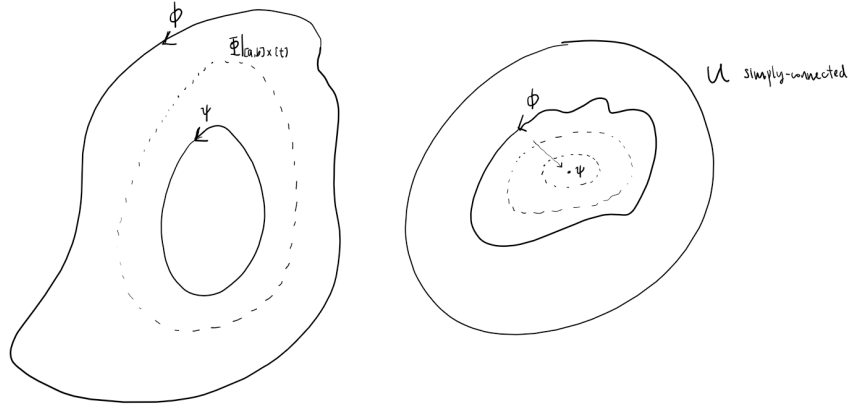
Let us recap some properties of holomorphic functions.

- (i) The Fundamental Theorem of Calculus (FTC), $\int_{\gamma} F' = F(\gamma(b)) - F(\gamma(a))$ where $\gamma : [a, b] \rightarrow U$.
- (ii) Cauchy's theorem for triangles, $\int_{\partial T} f = 0$ if $T \subseteq U$.
- (iii) Convex Cauchy, $\int_{\gamma} f = 0$ for all closed γ in a starlike domain.
- (iv) Taylor's theorem, f is infinitely differentiable.

Some criteria for holomorphicity include the following.

- (i) Converse of the FTC, if $\int_{\gamma} f = 0$ for all closed γ , then $f = F'$ for some holomorphic F .
- (ii) If U is starlike, the converse of FTC holds if $\int_{\partial T} f = 0$ for all $T \subseteq U$.
- (iii) Morera's theorem, if $\int_{\gamma} f = 0$ for all closed γ , then f is holomorphic on any domain.

We would like to expand Cauchy's theorem to all closed paths and all simply connected domains. To do this, we need to develop some more theory.



Let ϕ and ψ be closed paths in a domain U , say $\phi, \psi : [a, b] \rightarrow U$. We say ϕ and ψ are homotopic if there exists a continuous $\Phi : [a, b] \times [0, 1] \rightarrow U$ such that $\Phi|_{[a, b] \times \{0\}} = \phi$ and $\Phi|_{[a, b] \times \{1\}} = \psi$, and $\Phi|_{[a, b] \times \{t\}}$ is a closed path for all t . By definition, U is simply connected if every closed loop $\phi : [a, b] \rightarrow U$ is homotopic to a constant loop $\psi : [a, b] \rightarrow U$ with $\psi(t) = \psi(a)$ for all t .

Proposition 7.1

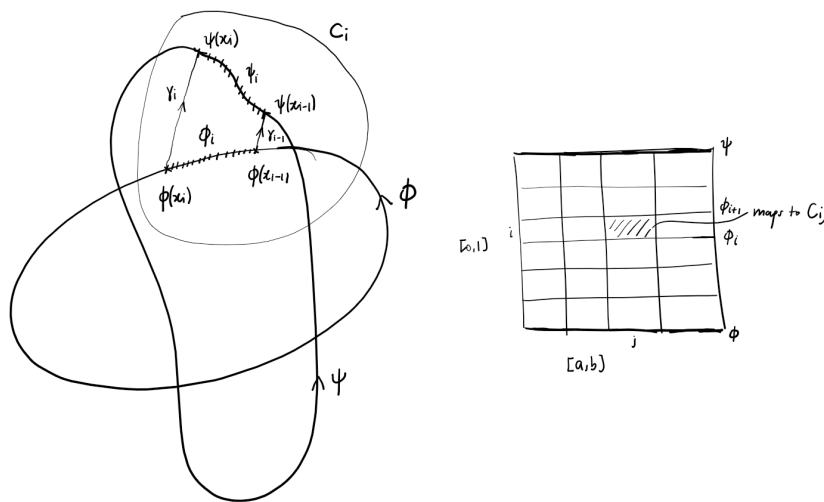
U is simply-connected if and only if whenever ϕ is a piecewise- C^1 closed path, there exists Φ a homotopy to the constant path such that $\Phi|_{[a, b] \times \{t\}}$ is a piecewise- C^1 closed path.

We will take this as a fact. In a domain, any two points are connected by a piecewise- C^1 path, this is the analogue for loops.

Proposition 7.2

Suppose U is a domain and ϕ, ψ are homotopic closed piecewise- C^1 paths (via a homotopy thorough such paths). If $f : U \rightarrow \mathbb{C}$ is holomorphic, then $\int_{\phi} f = \int_{\psi} f$.

Proof. Let $\phi, \psi : [a, b] \rightarrow U$ be closed paths. We say ϕ and ψ are elementary deformations of one another if there exist $a = x_0 < x_1 < \dots < x_n = b$ and convex open subsets $C_i \subseteq U$ for each $1 \leq i \leq n$ such that $\phi_i = \phi|_{[x_{i-1}, x_i]}$ and $\psi_i = \psi|_{[x_{i-1}, x_i]}$ have images inside C_i . Let $\gamma_i \subseteq C_i$ be the straight path from $\phi(x_i)$ to $\psi(x_i)$. Note that C_i is convex so γ_i lies wholly inside C_i . Let $\Gamma_i = \phi_i + \gamma_i - \psi_i - \gamma_{i-1}$ be a closed piecewise- C^1 loop inside C_i . Convex Cauchy implies that $\int_{\Gamma_i} f = 0$ so $\sum_{i=1}^n \int_{\Gamma_i} f = 0$. The γ_i all appear twice with opposite orientation, so $\int_{\phi} f = \int_{\psi} f$ if ϕ and ψ are elementary deformations of each other.



It now suffices to prove that if ϕ and ψ are homotopic, then there exist $\phi = \phi_0, \phi_1, \dots, \phi_N = \psi$ such that ϕ_{i+1} is an elementary deformation of ϕ_i . Let $\Phi : [a, b] \times [0, 1] \rightarrow U$ be the homotopy from $\phi = \Phi_0$ to $\psi = \Phi_1$. Since $[a, b] \times [0, 1]$ is closed and bounded, its image $\text{im}(\Phi) = \Phi([a, b] \times [0, 1])$ is closed and bounded. So, we can find $\varepsilon > 0$ such that $\text{distance}(\text{im}(\Phi), \mathbb{C} \setminus U) = \min\{|p - q| \mid p \in \text{im}(\Phi), q \in \mathbb{C} \setminus U\} = 2\varepsilon$. So, $B(\Phi(s, t), \varepsilon) \subseteq U$ for all $(s, t) \in [a, b] \times [0, 1]$. Now Φ is uniformly continuous, so there exists $\delta > 0$ such that $|(s, t) - (s', t')| < \delta$ implies $|\Phi(s, t) - \Phi(s', t')| < \varepsilon$.

Pick n such that $(1 + (b - a))/n < \delta$. Set $x_j = a + (b - a)j/n$ for $1 \leq j \leq n$ and $\phi_i = \Phi|_{[a, b] \times i/n} = \Phi_{i/n}$, so $\phi = \phi_0, \phi_1, \dots, \phi_n = \psi$. Let $C_{ij} = B(\Phi(x_j, i/n), \varepsilon) \subseteq U$ which is convex. Now if $s \in [x_{j-1}, x_j]$ and $t \in [(i-1)/n, i/n]$, then $\Phi(s, t) \in C_{ij}$. Hence, this shows that ϕ_{i+1} is an elementary deformation of ϕ_i . $\square$

Theorem 7.3 – Cauchy's theorem for simply connected domains

If $U \subseteq \mathbb{C}$ is a simply connected domain and $f : U \rightarrow \mathbb{C}$ is holomorphic, then $\int_\gamma f = 0$ for all closed piecewise- C^1 curves γ in U .

Proof. Follows from Proposition 7.2, by taking ψ to be a constant path. $\square$

8 Identity theorems

We would now discuss the zeroes of holomorphic functions. Suppose $f : B(a, \varepsilon) \rightarrow \mathbb{C}$ is holomorphic, so we know $f(z) = \sum_{n=0}^{\infty} c_n(z - a)^n$. If $f \not\equiv 0$, there exists a least m such that $c_m \neq 0$. If $m > 0$, we have $f(a) = 0$ and we can write $f(z) = (z - a)^m g(z)$ where g is holomorphic on $B(a, \varepsilon)$ with $g(a) \neq 0$. We say f has a zero of order m at a .

Proposition 8.1 – Principle of isolated zeroes

If $f : B(a, r) \rightarrow \mathbb{C}$ is holomorphic and $f \not\equiv 0$, then there exists $0 < \rho < r$ such that $f(z) \neq 0$ for all $z \in B(a, \rho) \setminus \{a\} =: B(a, \rho)^*$ where $B(a, \rho)^*$ is a punctured disc.

Proof. If $f(a) \neq 0$, the result is immediate by the continuity of f . If $f(a) = 0$, we said $f(z) = (z - a)^m g(z)$ where $g(a) \neq 0$. By the continuity of g , there exists ρ such that $g(z) \neq 0$ for all $z \in B(a, \rho)$. This same ρ works for the proposition. $\square$

Theorem 8.1 – Identity theorem

Let $U \subseteq \mathbb{C}$ be a domain and $f, g : U \rightarrow \mathbb{C}$ be holomorphic on U . Let $S = \{z \in U \mid f(z) = g(z)\}$. If S has an accumulation point in U (that is, there exists $w \in U$ such that $B(w, \varepsilon)^* \cap S \neq \emptyset$ for all $\varepsilon > 0$), then $f(z) = g(z)$ for all $z \in U$.

Proof. Let $h(z) = f(z) - g(z)$. h is holomorphic on U and vanishes on S . Suppose w is an accumulation point of S , so it is a non-isolated zero of h . By the principle of isolated zeroes, there exists $\varepsilon > 0$ such that $B(w, \varepsilon) \subseteq U$ and $h \equiv 0$ on $B(w, \varepsilon)$. Let $z \in U$ and fix a path $\gamma : [0, 1] \rightarrow U$ such that $\gamma(0) = w$ and $\gamma(1) = z$. Let

$$I = \{t \in [0, 1] \mid h^{(n)}(\gamma(t)) = 0 \text{ for all } n \geq 0\}.$$

By Taylor's theorem,

$$I = \{t \in [0, 1] \mid h \text{ vanishes on a small open neighbourhood of } \gamma(t)\}.$$

The first characterisation of I implies that it is closed. The second characterisation implies that it is open. Since $0 \in I \neq \emptyset$, we must have $I = [0, 1]$ as it is both open and closed. Hence, $h(z) = 0$ and $h \equiv 0$ on U . $\square$

- (i) If f is holomorphic on a domain U with $f \not\equiv 0$, then the zeroes of f cannot accumulate in U . However, they can still accumulate at a point not in U . Consider $U = \mathbb{C}^*$ and $f(z) = \sin(1/z)$. Then, $f(z) = 0$ when $z = 1/(k\pi)$ with $k \in \mathbb{Z}^+$. Hence, zeroes do accumulate at 0.
- (ii) An identity between entire functions holding on $\mathbb{R}$ implies that it also holds on $\mathbb{C}$. For example, we know $\sin^2 x + \cos^2 x = 1$ so $\sin^2 z + \cos^2 z - 1 \equiv 0$.

If U is a domain, we write $\overline{U}$ to denote the closure of U . That is, the smallest closed set in $\mathbb{C}$ containing U , or $U \cup \{\text{limits of sequences in } U\}$. We write $\partial U = \overline{U} \setminus U = \{z \in \mathbb{C} \mid B(z, \varepsilon) \cap U \neq \emptyset \text{ for all } \varepsilon > 0\} \setminus U$ to denote the boundary of U .

Corollary 8.1 – Global maximum principle

Let $U \subseteq \mathbb{C}$ be a bounded domain. Let $f : \overline{U} \rightarrow \mathbb{C}$ be continuous and holomorphic on U . Then, $|f|$ achieves its maximum on ∂U .

Proof. If $U \subseteq B(0, N)$, then $\overline{U} \subseteq \overline{B(0, N)}$, so $\overline{U}$ is closed and bounded. Thus, $|f| : \overline{U} \rightarrow \mathbb{R}$ does achieve its maximum M . Suppose $|f(a)| = M$ with $a \in U$. Pick $\varepsilon > 0$ such that $B(a, \varepsilon) \subseteq U$. By the local maximum principle, f is constant on $B(a, \varepsilon)$. Then, f and a constant function agree on $B(a, \varepsilon) \subseteq U$, so they agree on U by the identity theorem. Thus, f is constant and $|f|$ also achieves its maximum value M on ∂U . $\square$

9 Isolated singularities

Definition 9.1 – Analytic continuation

Let $U \subseteq U' \subseteq \mathbb{C}$ be domains and $f : U \rightarrow \mathbb{C}$ be holomorphic. A holomorphic function $h : U' \rightarrow \mathbb{C}$ such that $h|_U = f$ is an analytic continuation of f on U' .

This need not exist, but if it exists, then it is unique by the identity theorem.

Definition 9.2 – Natural boundary

If $f : U \rightarrow \mathbb{C}$ is holomorphic and admits no analytic continuation to any domain $U' \supsetneq U$, then we say that ∂U is the natural boundary of f .

Example 9.1

- (i) Let $f(z) = \sum_{n=0}^{\infty} z^n$ with radius of convergence 1. This defines a holomorphic function on $\mathbb{D} = B(0, 1)$. It has an analytic continuation $1/(1-z)$ defined on $\mathbb{C} \setminus \{1\}$.
- (ii) On example sheet 2, we see that $f(z) = \sum_{n=0}^{\infty} z^{(2^n)}$ defines a holomorphic function on $\mathbb{D}$ with natural boundary $\partial \mathbb{D} = S^1$.

Recall Corollary 7.3. We needed f to be continuous on U and holomorphic on $U \setminus S$, where S is a finite set. What happens if f is not continuous at S ?

Proposition 9.1 – Removal of singularities

Let U be a domain and $z_0 \in U$. Let $f : U \setminus \{z_0\} \rightarrow \mathbb{C}$ be holomorphic and suppose f is bounded near z_0 . Then, there exists $a \in \mathbb{C}$ such that $f(z) \rightarrow a$ as $z \rightarrow z_0$ and

$$g(z) = \begin{cases} f(z) & \text{if } z \in U \setminus \{z_0\} \\ a & \text{if } z = z_0 \end{cases}$$

is holomorphic on U .

Proof. By the hypothesis, there exist $\delta > 0$ and $M > 0$ such that $|f(z)| < M$ for all $z \in B(z_0, \delta)^*$. Define $h : U \rightarrow \mathbb{C}$ by

$$h(z) = \begin{cases} (z - z_0)^2 f(z) & \text{if } z \neq z_0 \\ 0 & \text{if } z = z_0. \end{cases}$$

Then, h is complex differentiable on U . This is clear for $z \neq z_0$. Near z_0 , we have

$$\left| \frac{h(z) - h(z_0)}{z - z_0} \right| \leq M|z - z_0|$$

so its limit exists and is zero. Thus, h is holomorphic, so it has a Taylor series convergent near z_0 and we can write $h(z) = \sum_{n=0}^{\infty} a_n(z - z_0)^n$ on a ball $B(z_0, \varepsilon)$. We have $h(z_0) = h'(z_0) = 0$ so $a_0 = a_1 = 0$. Let $g(z) = \sum_{n=0}^{\infty} a_{n+2}(z - z_0)^n$ which is also convergent on $B(z_0, \varepsilon)$. By construction, g agrees with f on $B(z_0, \varepsilon)^*$, so extends to U by analytic continuation. Indeed, $\lim_{z \rightarrow z_0} f(z) = a_2$ and we are done. $\square$

Proposition 9.2

Let U be a domain and $z_0 \in U$. Let $f : U \setminus \{z_0\} \rightarrow \mathbb{C}$ be holomorphic and suppose $|f(z)| \rightarrow \infty$ as $z \rightarrow z_0$. Then, there is a holomorphic function $g : U \rightarrow \mathbb{C}$ with $g(z_0) \neq 0$, and $k \geq 1$ such that

$$f(z) = \frac{g(z)}{(z - z_0)^k}$$

for all $z \neq z_0$. Both g and k are unique.

Proof. Pick $\delta > 0$ such that $|f(z)| \geq 1$ for all $z \in B(z_0, \delta)^*$. Let

$$h(z) = \begin{cases} 1/f(z) & \text{if } z \in B(z_0, \delta)^* \\ 0 & \text{if } z = z_0. \end{cases}$$

Proposition 9.1 says that h is holomorphic on $B(z_0, \delta)$. From our discussion on the zeroes of holomorphic functions, there exists a unique $k \geq 1$ such that h has a zero of order k at z_0 . That is, $h(z_0) = h'(z_0) = \dots = h^{(k-1)}(z_0) = 0$. Using a Taylor series for h , we have $h(z) = (z - z_0)^k l(z)$ where $l : B(z_0, \delta) \rightarrow \mathbb{C}$ is holomorphic and $l(z_0) \neq 0$. l is continuous so there exists $\varepsilon > 0$ with $0 < \varepsilon < \delta$ such that $l(z) \neq 0$ for all $z \in B(z_0, \varepsilon)$. Let $g(z) = 1/l(z)$ for $z \in B(z_0, \varepsilon)$. Then, g is holomorphic on $B(z_0, \varepsilon)$ and $g(z) = (z - z_0)^k f(z)$ for all $z \in B(z_0, \varepsilon)^*$. The final expression shows that g has an analytic continuation to U , by the identity theorem. $\square$

Definition 9.3 – Isolated singularities

Let U be a domain and $z_0 \in U$. Let $f : U \setminus \{z_0\} \rightarrow \mathbb{C}$ be holomorphic. We say that f has an isolated singularity at z_0 and that this singularity is

- (i) removable if $|f|$ is bounded near z_0
- (ii) a pole of order k if $|f| \rightarrow \infty$ near z_0 and $f(z) = g(z)/(z - z_0)^k$ with g holomorphic and $g(z_0) \neq 0$
- (iii) essential otherwise, in which case $|f|$ has no limit in $\mathbb{R} \cup \{\infty\}$ near z_0

We showed that for case (i), f has a limit as z_0 . Hence, (iii) holds if and only if $|f|$ has no limit in $\mathbb{R} \cup \{\infty\}$ at z_0 . For case (ii), we saw that there exist a unique $k \geq 1$ and holomorphic g on U with $g(z_0) \neq 0$ such that $f(z) = g(z)/(z - z_0)^k$. We say that the pole has order k .

Example 9.2

- (i) Consider $f(z) = \sin(z)/z$ on $\mathbb{C}^*$. We know that $\sin(z) = z - z^3/3! + z^5/5! - \dots$ so $\sin(z)/z = 1 - z^2/6 + z^4/120 - \dots$ is holomorphic near $z = 0$, so the singularity of f is removable.
- (ii) $f(z) = 1/z^3$ has an order 3 pole at $z = 0$.
- (iii) Consider $f(z) = \exp(1/z)$ on $\mathbb{C}^*$. Recall that $\exp$ maps a horizontal strip of width 2π onto $\mathbb{C}^*$. But $z \mapsto 1/z : B(0, \varepsilon)^* \rightarrow \mathbb{C} \setminus \overline{B(0, 1/\varepsilon)}$ and the right hand side contains a horizontal strip of width 2π . So, f maps every punctured neighbourhood of 0 to $\mathbb{C}^*$. Hence, $|f|$ has no limit at 0 and the singularity is essential.

Theorem 9.1 – Casorati-Weierstrass

Let U be a domain and $z_0 \in U$. If $f : U \setminus \{z_0\} \rightarrow \mathbb{C}$ is holomorphic with an essential singularity at z_0 , then for all $w \in \mathbb{C}$, there exists a sequence $z_n \rightarrow z_0$ such that $f(z_n) \rightarrow w$.

Proof. On example sheet 2. $\square$

A stronger statement, which we will not prove, is Picard's theorem.

Theorem 9.2 – Picard's theorem

Let U be a domain and $z_0 \in U$. If $f : U \setminus \{z_0\} \rightarrow \mathbb{C}$ has an isolated essential singularity at z_0 , then there exists $v \in \mathbb{C}$ such that for all $\varepsilon > 0$, we have $f(B(z_0, \varepsilon)^*) = \mathbb{C} \setminus \{v\}$.

This suggests that $\mathbb{C} \setminus \{\text{one point}\}$ is topologically or analytically very different from $\mathbb{C} \setminus \{\text{two or more points}\}$.

Let $f : B(z_0, r)^* \rightarrow \mathbb{C}$ be holomorphic with a pole at z_0 . Since $|f| \rightarrow \infty$ as $z \rightarrow z_0$, f defines a map $B(z_0, r) \rightarrow \mathbb{C} \cup \{\infty\}$ that sends z_0 to ∞ .

Definition 9.4 – Meromorphic

Let U be a domain and $S \subseteq U$ be finite. If $f : U \setminus S \rightarrow \mathbb{C}$ is holomorphic with poles or removable singularities at points of S , we say that f is meromorphic on U .

Example 9.3

If p and q are polynomials with coefficients in $\mathbb{C}$, then p/q is meromorphic on $\mathbb{C}$.

10 Laurent series

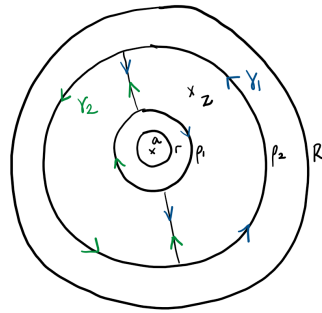
The next result governs holomorphic functions on $B(a, r)^*$ or more generally on an annulus.

Theorem 10.1 – Laurent series

Let $A = \{z \in \mathbb{C} \mid r < |z - a| < R\}$ where $0 \leq r < R \leq \infty$. If $f : A \rightarrow \mathbb{C}$ is holomorphic, then the following holds.

- (i) f has a unique convergent power series expansion $f(z) = \sum_{n=-\infty}^{\infty} c_n(z - a)^n$ convergent on A .
- (ii) If $r < \rho' \leq \rho < R$, the series converges uniformly on $\{\rho' \leq |z - a| \leq \rho\}$.
- (iii) For $r < \rho < R$, we have $c_n = \frac{1}{2\pi i} \int_{\partial B(a, \rho)} \frac{f(w)}{(w - a)^{n+1}} dw$.

Proof.



Fix $z \in A$. Pick ρ_1 and ρ_2 such that $r < \rho_1 < |z - a| < \rho_2 < R$. Let γ_1 and γ_2 be piecewise- C^1 curves as shown. Cauchy's integral formula gives

$$f(z) = \frac{1}{2\pi i} \int_{\gamma_1} \frac{f(w)}{w - z} dw.$$

Cauchy's theorem gives

$$0 = \frac{1}{2\pi i} \int_{\gamma_2} \frac{f(w)}{w - z} dw.$$

Adding these, and noting that the line segments in γ_1 and γ_2 appear with opposite orientation, we have

$$f(z) = \frac{1}{2\pi i} \left[\int_{\partial B(a, \rho_2)} \frac{f(w)}{w - z} dw - \int_{\partial B(a, \rho_1)} \frac{f(w)}{w - z} dw \right].$$

As in the proof for Taylor's theorem, write

$$\frac{1}{w-z} = \frac{1}{w-a} \frac{1}{1 - \frac{z-a}{w-a}} = \sum_{n=0}^{\infty} \frac{(z-a)^n}{(w-a)^{n+1}}$$

which converges uniformly on $|w-a| = \rho_2$ since $|z-a| < \rho_2$. We will use this for the first integral. Also,

$$-\frac{1}{w-z} = \frac{1}{z-a} \frac{1}{1 - \frac{w-a}{z-a}} = \sum_{m=1}^{\infty} \frac{(w-a)^{m-1}}{(z-a)^m}$$

which converges uniformly on $|w-a| = \rho_1$ since $|z-a| > \rho_1$. We will use this for the second integral. In both cases, uniform convergence lets us swap $\sum$ and $\int$. Thus,

$$f(z) = \sum_{n=0}^{\infty} \left(\frac{1}{2\pi i} \int_{\partial B(a, \rho_2)} \frac{f(w)}{(w-a)^{n+1}} dw \right) (z-a)^n + \sum_{m=1}^{\infty} \left(\frac{1}{2\pi i} \int_{\partial B(a, \rho_1)} \frac{f(w)}{(w-a)^{-m+1}} \right) (z-a)^{-m}.$$

Let $n = -m$ in the second series. Replace both contours $\partial B(a, \rho_2)$ and $\partial B(a, \rho_1)$ by a common contour $|w-a| = \rho$ using homotopic invariance of the integral. This gives us the Laurent series.

It remains to show uniqueness. If $f(z) = \sum_{n=-\infty}^{\infty} c_n(z-a)^n = \sum_{m=-\infty}^{\infty} b_m(z-a)^m$, then

$$\begin{aligned} 2\pi i c_k &= \int_{\partial B(a, \rho)} \frac{f(w)}{(w-a)^{k+1}} dw \\ &= \int_{\partial B(a, \rho)} \sum_{m=-\infty}^{\infty} b_m (w-a)^{m-k-1} dw \\ &= \sum_{m=-\infty}^{\infty} b_m \int_{\partial B(a, \rho)} (w-a)^{m-k-1} dw \\ &= 2\pi i b_k \end{aligned}$$

as all the other integrals vanish. □

In Taylor's theorem, $f(z) = \sum_{n=0}^{\infty} c_n(z-a)^n$ if f is holomorphic on $B(a, R)$, and $c_n = f^{(n)}(a)/n!$. But in Laurent's theorem, where f is holomorphic on $B(a, R)^*$, there is no general formula for c_n .

Corollary 10.1

If $r = 0$ in Theorem 10.1, then $A = B(a, R)^*$ and f has an isolated singularity at a .

- (i) a is removable if and only if $c_n = 0$ for all $n < 0$.
- (ii) a is a pole of order k if and only if $c_{-k} \neq 0$ and $c_n = 0$ for all $n < -k$.
- (iii) a is essential if and only if $c_n \neq 0$ for infinitely many negative n .

Proof. (i) and (ii) are immediate from our previous characterisations. (iii) is by exclusion. □

11 Series expansions

Definition 11.1 – Laurent series and its convergence

For $a \in \mathbb{C}$, a Laurent series centred at a is a sum $\sum_{n=-\infty}^{\infty} c_n(z-a)^n$. By definition, this converges at $z \in \mathbb{C}$ if both $\sum_{n=0}^{\infty} c_n(z-a)^n$ and $\sum_{n=-\infty}^{-1} c_n(z-a)^n = \sum_{n=1}^{\infty} c_{-n}(z-a)^{-n}$ converges.

From our results on power series, there exists $R > 0$ such that $\sum_{n=0}^{\infty} c_n(z-a)^n$ converges absolutely on

$\{|z - a| < R\}$ and converges uniformly on $\{|z - a| \leq \rho\}$ with $\rho < R$. Similarly, if we set $u = 1/(z - a)$, then $\sum_{n=1}^{\infty} c_{-n}(z - a)^{-n} = \sum_{n=1}^{\infty} c_{-n}u^n$ has a radius of convergence $R' > 0$ such that it converges absolutely if $|u| < R'$. That is, if $|z - a| > 1/R' =: r$.

If $r < R$, the Laurent series converges absolutely on $A = \{r < |z - a| < R\}$ and converges uniformly on $\{\rho \leq |z - a| \leq \rho'\}$ with $r < \rho \leq \rho' < R$. Theorem 10.1 says that if f is holomorphic on an annulus A , it admits a unique Laurent series expansion centred at a , the centre of A . In our proof, we explicitly constructed power series f_{out} and f_{in} such that $f = f_{\text{out}} + f_{\text{in}}$, with f_{out} holomorphic on $|z - a| > R$ and f_{in} holomorphic on $|z - a| < r$.

Definition 11.2 – Principal part

If $f(z) = \sum_{n=-\infty}^{\infty} c_n(z - a)^n$ on an annulus A , the sum $\sum_{n=-\infty}^{-1} c_n(z - a)^n$ is said to be the principal part of f , written $f_{\text{prin}}(z)$.

If f is holomorphic on $B(a, R)^*$ so $r = 0$, then f_{prin} defines a holomorphic function on $\mathbb{C} \setminus \{a\}$, which vanishes at infinity since the power series in $u = 1/(z - a)$ has no constant term. Also, $f - f_{\text{prin}}$ is holomorphic near a .

Example 11.1

Consider $\csc z = 1/(\sin z)$ which is holomorphic except at $z = k\pi$ where $k \in \mathbb{Z}$. So, it has a Laurent series on $\{0 < |z| < \pi\}$. Now $\sin z = z(1 - z^2/6 + \mathcal{O}(z^4))$ so

$$\csc z = \frac{1}{z} \left[1 - \frac{z^2}{6} + \mathcal{O}(z^4) \right]^{-1} = \frac{1}{z} \left[1 + \frac{z^2}{6} + \mathcal{O}(z^4) \right].$$

So, $c_{-1} = 1$ and $\csc$ has a simple pole at 0.

Example 11.2

We want to prove that if $z \in \mathbb{C} \setminus \mathbb{Z}$, then

$$\sum_{n \in \mathbb{Z}} \frac{1}{(z - n)^2} = \frac{\pi^2}{\sin^2(\pi z)}.$$

For $z \in \mathbb{C} \setminus \mathbb{Z}$, set $f(z) = \sum_{n=-\infty}^{\infty} 1/(z - n)^2$. If $w \in \mathbb{C} \setminus \mathbb{Z}$, we can pick $r > 0$ such that $|w - n| > 2r$ for all $n \in \mathbb{Z}$. Then for all $z \in B(w, r)$, we have

$$\begin{aligned} |z - n| &\geq \max\{r, n - |w| + r\} \\ \frac{1}{|z - n|^2} &\leq \min \left\{ \frac{1}{r^2}, \frac{1}{(n - |w| + r)^2} \right\} =: M_n. \end{aligned}$$

$\sum_{n \geq 0} M_n$ converges by comparison to $\sum 1/n^2$, so by the Weierstrass M-test, f converges uniformly on $B(w, r)$. Hence, f is holomorphic on $\mathbb{C} \setminus \mathbb{Z}$.

Note that $f(z + 1) = f(z)$ and f has a double pole at all $n \in \mathbb{Z}$. Let $g(z) = (\pi / \sin(\pi z))^2$, then g is also $\mathbb{Z}$ -periodic and satisfies $g(z + 1) = g(z)$. From our power series for $\csc$, g also has double poles at $n \in \mathbb{Z}$. At $z = n$, we have $f_{\text{prin}}(z) = 1/(z - n)^2 = g_{\text{prin}}(z)$ by checking that $\lim_{z \rightarrow 0} z^2 g(z) = 1$ and using periodicity. Since $f(z) - g(z)$ is holomorphic on $\mathbb{C} \setminus \mathbb{Z}$ and has trivial principal part at each $n \in \mathbb{Z}$, $h(z) = f(z) - g(z)$ is entire. We want to show $h \equiv 0$. By Liouville's theorem, it suffices to prove that h is bounded and $h \rightarrow 0$ near infinity.

By periodicity, we can consider h only on $-1/2 \leq \text{Re}(z) \leq 1/2$. Write $z = x + iy$ so $|x| \leq 1/2$. If $y > 0$,

then

$$|g(z)| \leq \frac{4\pi^2}{|e^{\pi y} - e^{-\pi y}|^2} \rightarrow 0$$

$$|f(z)| \leq \sum_{n \in \mathbb{Z}} \frac{1}{|x + iy - n|^2} \leq \frac{1}{y^2} + 2 \sum_{n=1}^{\infty} \frac{1}{(n - 1/2)^2 + y^2} \rightarrow 0$$

as $y \rightarrow \infty$. So, there exists $Y > 0$ such that $|f|$ and $|g|$ are bounded by 1 when $y > Y$, and it is bounded inside $y \leq Y$ by continuity.

12 Winding and residue

Our next topic will be the theory of winding numbers. We know that $\int_{\partial \mathbb{D}} 1/z \, dz = 2\pi i$. If $\gamma : [0, 1] \rightarrow \mathbb{C}$ is the curve $t \mapsto e^{4\pi i t}$, then $\int_{\gamma} 1/z \, dz = 4\pi i$. The integral detects that one curve winds more around 0 than the other.

Lemma 12.1

Let $\gamma : [a, b] \rightarrow \mathbb{C} \setminus \{w\}$ be a path such that $w \notin \text{im}(\gamma)$. There are continuous functions $r, \theta : [a, b] \rightarrow \mathbb{R}$ such that $\gamma(t) = w + r(t)e^{i\theta(t)}$.

Proof. By translating, we can assume $w = 0$. Set $r(t) = |\gamma(t)|$. Replacing γ by γ/r , we can assume $|\gamma(t)| = 1$ for all t without loss of generality. If $\gamma(t) \subseteq \mathbb{C} \setminus \mathbb{R}_{\leq 0}$, we can set $\theta(t) = \text{Arg}(\gamma(t)) \in (-\pi, \pi)$. More generally, if $\gamma(t) = \mathbb{C} \setminus \{se^{i\alpha} \mid s \in \mathbb{R}_{\leq 0}\}$, then there exists a branch of logarithm on the image of γ , and we can define θ . γ is continuous on $[a, b]$ so is uniformly continuous, so there exists $\delta > 0$ such that $|s - t| < \delta$ implies $|\gamma(s) - \gamma(t)| < 2$ and $\gamma(s), \gamma(t)$ are not antipodal. Write $a = a_0 < a_1 < \dots < a_N = b$ such that $|a_{i-1} - a_i| < \delta$, so $\gamma|_{[a_{i-1}, a_i]}$ has image in a slit plane. So, for all n , there exists $\theta_n : [a_{n-1}, a_n] \rightarrow \mathbb{R}$ such that $\gamma(t) = e^{i\theta_n(t)}$ on $[a_{n-1}, a_n]$. Note that $\theta_{n-1}(a_{n-1}) = \theta_n(a_{n-1}) + b_n$ for some $b_n \in 2\pi\mathbb{Z}$. Redefine $\theta_n^{\text{new}} = \theta_n^{\text{old}} + b_n$ iteratively, then θ is continuous. $\square$

$\theta(t)$ is not unique in this construction. If $\gamma(t) = w + r(t)e^{i\theta(t)} = w + r(t)e^{i\tilde{\theta}(t)}$ with θ and $\tilde{\theta}$ continuous, then $(\theta - \tilde{\theta}) : [a, b] \rightarrow 2\pi\mathbb{Z}$ is constant. So, $\tilde{\theta}(t) = \theta(t) + 2\pi k$ for some fixed $k \in \mathbb{Z}$.

Definition 12.1 – Winding number

Suppose $\gamma : [a, b] \rightarrow \mathbb{C} \setminus \{w\}$ is a continuous closed path. The winding number $I(\gamma, w)$ of γ about w is

$$\frac{\theta(b) - \theta(a)}{2\pi} \in \mathbb{Z}$$

where $\gamma(t) = w + r(t)e^{i\theta(t)}$.

This is well-defined as it is independent of the choice of $\theta(t)$.

Lemma 12.2

If $\gamma : [a, b] \rightarrow \mathbb{C} \setminus \{w\}$ is a piecewise- C^1 closed path, then

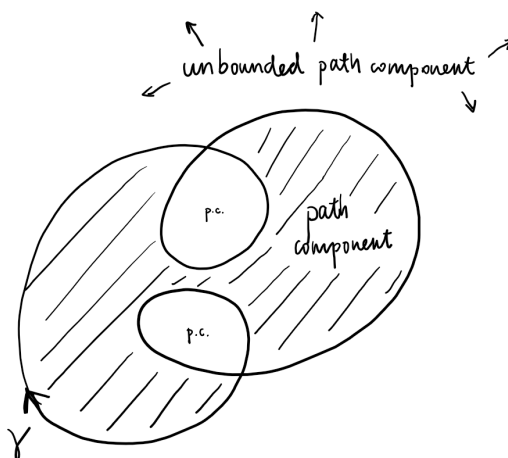
$$I(\gamma, w) = \frac{1}{2\pi i} \int_{\gamma} \frac{1}{z - w} \, dz.$$

Proof. When γ is piecewise- C^1 , both $r(t)$ and $\theta(t)$ are piecewise- C^1 . Then, by writing $\gamma = w + r(t)e^{i\theta(t)}$,

$$\begin{aligned} \int_{\gamma} \frac{1}{z-w} dz &= \int_a^b \frac{\gamma'(t)}{\gamma(t)-w} dt \\ &= \int_a^b \left[\frac{r'(t)}{r(t)} + i\theta'(t) \right] dt \\ &= [\ln(r(t)) + i\theta(t)]_a^b \\ &= i(\theta(b) - \theta(a)) \\ &= 2\pi i I(\gamma, w). \end{aligned}$$

□

If $\gamma(t)$ is a loop, we can consider $I(\gamma, w)$ as a function of $w \notin \text{im}(\gamma)$. In general, $\mathbb{C} \setminus \text{im}(\gamma)$ may have many path components. Note that there is exactly one unbounded path component, since $\text{im}(\gamma)$ is a closed bounded set.



Proposition 12.1

- (i) $I(\gamma, w)$ is constant as a function of w on path components of $\mathbb{C} \setminus \text{im}(\gamma)$.
- (ii) On the unbounded path component, $I(\gamma, w) = 0$.

Proof. For (i), note $I(\gamma, w) = (1/(2\pi i)) \int_{\gamma} 1/(z-w) dz$ is continuous on w . Thus, $I(\gamma, w) : \mathbb{C} \setminus \text{im}(\gamma) \rightarrow \mathbb{Z}$ is continuous, and so constant on any path, and hence constant on a path component.

For (ii), suppose $\text{im}(\gamma) \subseteq B(0, R)$ for some large R . Pick w such that $|w| > 2R$. Then,

$$|I(\gamma, w)| = \frac{1}{2\pi} \left| \int_{\gamma} \frac{dz}{z-w} \right| \leq \frac{\text{length}(\gamma)}{2\pi} \frac{1}{R}.$$

As $R \rightarrow \infty$, the right hand side is < 1 . Since $I(\gamma, w) \in \mathbb{Z}$, it must be zero.

□

Let us go back to a holomorphic function $f : B(a, r)^* \rightarrow \mathbb{C}$ with an isolated singularity at a , and with a Laurent series $f(z) = \sum_{n=-\infty}^{\infty} c_n(z-a)^n$.

Definition 12.2 – Residue

c_{-1} in the Laurent series of f at an isolated singularity a is called the residue of f at a , written $\text{res}_f(a)$.

If $0 < \rho < r$, we have $\int_{\partial B(a, \rho)} f(z) dz = 2\pi i c_{-1} = 2\pi i \text{res}_f(a)$.

Theorem 12.1 – Cauchy’s residue theorem

Let U be a simply connected domain. Let $Z = \{z_1, \dots, z_k\} \subseteq U$ and $f : U \setminus Z \rightarrow \mathbb{C}$ be holomorphic. Let $\gamma : [a, b] \rightarrow U \setminus Z$ be a piecewise- C^1 closed path. Then,

$$\frac{1}{2\pi i} \int_{\gamma} f(z) dz = \sum_{j=1}^k I(\gamma, z_j) \operatorname{res}_f(z_j).$$

Proof. f has Laurent series $\sum_{n=-\infty}^{\infty} c_n^{(j)} (z - z_j)^n$ at z_j which is convergent in some $B(z_j, \rho_j)^* \subseteq U$. Let $g_j(z) = \sum_{n=-\infty}^{-1} c_n^{(j)} (z - z_j)^n$ be the principal part of this series. This is holomorphic on $\mathbb{C} \setminus \{z_j\}$ and in particular on $U \setminus \{z_j\}$. Consider $f - g_1 - \dots - g_k$ which is holomorphic on $U \setminus Z$, with removable singularities at points in Z , so this defines a holomorphic function on the whole of U . Cauchy’s theorem for simply connected domains then says $\int_{\gamma} (f - g_1 - \dots - g_k) = 0$. Thus,

$$\int_{\gamma} f = \sum_{j=1}^k \int_{\gamma} g_j(z) dz = 2\pi i \sum_{j=1}^k I(\gamma, z_j) c_{-1}^{(j)} = 2\pi i \sum_{j=1}^k I(\gamma, z_j) \operatorname{res}_f(z_j).$$

□

- (i) This generalises both Cauchy’s theorem (f had only removable singularities) and the Cauchy integral formula (by applying the residue theorem to $\hat{f}(z) = f(z)/(z - w)$ and $\gamma = \partial B(w, \rho)$).
- (ii) We say a piecewise- C^1 closed curve γ bounds a domain D in U if

$$I(\gamma, w) = \begin{cases} 1 & \text{if } w \in D \\ 0 & \text{if } w \notin D \end{cases}$$

for $w \in U \setminus \operatorname{im}(\gamma)$. If γ is simple and U is simply connected, then γ bounds the domain it encloses.

Corollary 12.1

If f is meromorphic on U and has no poles on γ , and γ bounds a domain D in U , then

$$\int_{\gamma} f = 2\pi i \sum_{\text{poles } w \in D} \operatorname{res}_f(w).$$

In this setting, f has only finitely many poles in D .

Lemma 12.3

Let $f : U \rightarrow \mathbb{C}$ be meromorphic with a pole at $a \in U$.

- (i) If the pole is simple (of order one), then $\operatorname{res}_f(a) = \lim_{z \rightarrow a} (z - a)f(z)$.
- (ii) If $f(z) = g(z)/h(z)$ where g and h are holomorphic at a with $g(a) \neq 0$, and h has a simple zero at a , then $\operatorname{res}_f(a) = g(a)/h'(a)$.
- (iii) If $f(z) = g(z)/(z - a)^k$ with g holomorphic near a , then $\operatorname{res}_f(a) = g^{(k-1)}(a)/(k-1)!$.

Proof. For (i), $f(z) = c_{-1}/(z - a) + c_0 + c_1(z - a) + \dots$ so clearly $c_{-1} = \lim_{z \rightarrow a} (z - a)f(z)$.

For (ii), h has a simple zero so $1/h$ has a simple pole. Using (i),

$$\operatorname{res}_f(a) = \lim_{z \rightarrow a} (z-a) \frac{g(z)}{h(z)} = g(a) \lim_{z \rightarrow a} \frac{z-a}{h(z)-h(a)} = \frac{g(a)}{h'(a)}$$

since $h(a) = 0$.

For (iii), we want the coefficient of $(z-a)^{k-1}$ in the Taylor series for the holomorphic function g . We arrive at the result by applying Taylor's theorem. $\square$

13 Computations

Example 13.1 $-\int_0^\infty \frac{1}{1+x^4} dx = \frac{\pi}{2\sqrt{2}}$

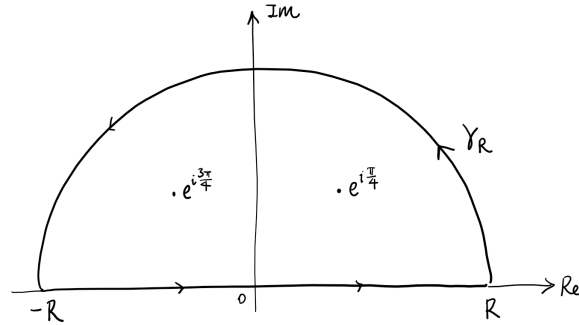
$f(z) = 1/(1+z^4)$ is meromorphic on $\mathbb{C}$. Consider the path γ_R which is the real axis from $-R$ to R union with the semicircle of radius R with $R > 2$. γ_R encloses two simple poles of f which are $p_1 = e^{i\pi/4}$ and $p_2 = e^{i3\pi/4}$. Note that $I(\gamma_R, p_i) = 1$. We have

$$\int_{\gamma_R} f(z) dz = \underbrace{\int_{-R}^R \frac{1}{1+x^4} dx}_{= 2 \int_0^\infty \frac{dx}{1+x^4} \text{ as } R \rightarrow \infty} + \underbrace{\int_0^\pi \frac{iRe^{i\theta}}{1+R^4e^{4i\theta}} d\theta}_{\mathcal{O}(1/R^3) \text{ as } R \rightarrow \infty}.$$

By the residue theorem and Lemma 12.3 (ii),

$$\int_{\gamma_R} f(z) dz = 2\pi i (\operatorname{res}_f(p_1) + \operatorname{res}_f(p_2)) = \frac{1}{4p_1^3} + \frac{1}{4p_2^3} = -\frac{1}{4}p_1 - \frac{1}{4}p_2.$$

Rearranging gives the result.



Example 13.2 $-\int_{\mathbb{R}} \frac{\cos(x)}{1+x+x^2} dx = \frac{2\pi}{\sqrt{3}} \cos(1/2)e^{-\sqrt{3}/2}$

At $z = iR$, $\cos(z) \approx e^R$ so we cannot easily bound $\cos(z)/(1+z+z^2)$ on γ_R as in Example 13.1. Let $f(z) = e^{iz}/(1+z+z^2)$ which has poles at zeroes of $1+z+z^2$. On the semicircle,

$$\left| \int_0^\pi f(Re^{i\theta}) iRe^{i\theta} d\theta \right| = \int_0^\pi \frac{Re^{-R\sin\theta}}{|R^2e^{2i\theta} + Re^{i\theta} + 1|} d\theta.$$

The integrand over the semicircle is $\mathcal{O}(1/R)$ and so tends to zero as $R \rightarrow \infty$. Note that the pole ω of f

satisfies $\omega^3 = 1$ and $I(\gamma_R, \omega) = 1$. Thus,

$$\int_{\mathbb{R}} \frac{\cos(x)}{1+x+x^2} dx = \operatorname{Re} \int_{\mathbb{R}} f(z) dz = \operatorname{Re}[2\pi i \operatorname{res}_f(\omega)]$$

which gives the result.

Lemma 13.1

Let $f : B(a, r)^* \rightarrow \mathbb{C}$ be holomorphic with a simple pole at a . For $0 < \varepsilon < r$, let $\gamma_\varepsilon : [\alpha, \beta] \rightarrow \mathbb{C}$ be defined by $t \mapsto a + \varepsilon e^{it}$. Then,

$$\lim_{\varepsilon \rightarrow 0} \int_{\gamma_\varepsilon} f(z) dz = (\beta - \alpha) i \operatorname{res}_f(a).$$

Proof. As f has a simple pole at a , we can write $f(z) = c/(z - a) + g(z)$ where g is holomorphic. g is continuous and locally bounded so

$$\left| \int_{\gamma_\varepsilon} g \right| \leq (\beta - \alpha) \varepsilon \sup_{z \in \gamma_\varepsilon} |g(z)| \rightarrow 0$$

as $\varepsilon \rightarrow 0$. But

$$\lim_{\varepsilon \rightarrow 0} \int_{\gamma_\varepsilon} \frac{c}{z - a} dz = c \lim_{\varepsilon \rightarrow 0} \int_\alpha^\beta \frac{1}{\varepsilon e^{it}} i \varepsilon e^{it} dt = (\beta - \alpha) ic.$$

Since $c = \operatorname{res}_f(a)$, we are done. $\square$

Lemma 13.2 – Jordan's lemma

If f is holomorphic on $\{|z| \geq r\}$ for some r , and that $zf(z)$ is bounded for large $|z|$, then for all $\alpha > 0$, we have

$$\int_{\gamma_R} f(z) e^{i\alpha z} dz \rightarrow 0$$

as $R \rightarrow \infty$, where $\gamma_R : [0, \pi] \rightarrow \mathbb{C}$ is the semicircular arc defined by $t \mapsto Re^{it}$.

Proof. For large $|z|$, we have $|f(z)| \leq C/|z|$ for some constant C . If $z = Re^{it}$, then $|e^{i\alpha z}| = e^{-R\alpha \sin t}$ and $\sin(t)/t \geq 2/\pi$ on $t \in [0, \pi/2]$. So,

$$|e^{i\alpha z}| \leq \begin{cases} e^{-R\alpha(2t/\pi)} & \text{on } 0 \leq t \leq \pi/2 \\ e^{-R\alpha(2(\pi-t)/\pi)} & \text{on } \pi/2 \leq t \leq \pi. \end{cases}$$

On the first half of γ_R , we have

$$\left| \int_0^{\pi/2} e^{i\alpha Re^{it}} f(Re^{it}) i Re^{it} dt \right| \leq \int_0^{\pi/2} e^{-2\alpha Rt/\pi} C dt = \frac{C\pi}{2\alpha R} (1 - e^{-\alpha R}) \rightarrow 0$$

as $R \rightarrow \infty$. The estimate for the second half is analogous and we are done. $\square$

Example 13.3 – $\int_0^\infty \frac{\sin(x)}{x} dx = \frac{\pi}{2}$

If $\hat{f}(z) = \sin(z)/z$, the singularity at 0 is removable, but is $\mathcal{O}(e^R/R)$ on γ_R . If $f(z) = e^{iz}/z$, we have a simple zero at 0, so to apply the residue theorem, we need a contour not passing through zero. We know $\int_{\gamma_{R,\varepsilon}} f(z) dz = 0$ for all $R, \varepsilon > 0$ since it does not pass through any singularities. Lemma 13.1 gives

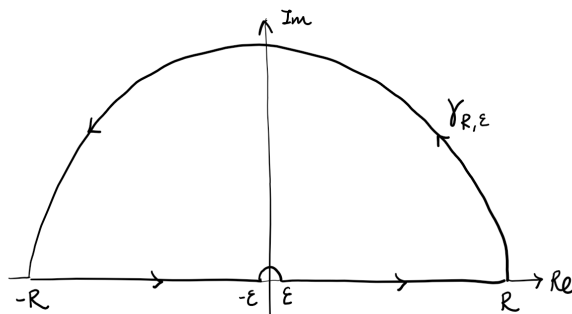
$\int_{\varepsilon\text{-semicircle}} e^{iz}/z dz = -i\pi$. Lemma 13.2 gives $\int_{R\text{-semicircle}} e^{iz}/z dz \rightarrow 0$ as $R \rightarrow \infty$. Hence,

$$\int_0^\infty \frac{\sin(x)}{x} dx = \frac{1}{2i} \int_0^\infty \frac{e^{ix} - e^{-ix}}{x} dx = \frac{1}{2i} \int_0^\infty \frac{e^{ix}}{x} dx - \frac{1}{2i} \int_{-\infty}^0 \frac{e^{ix}}{x} dx$$

and

$$0 = \int_{\gamma_{R,\varepsilon}} f(z) dz \rightarrow -i\pi + \int_{\mathbb{R}} \frac{e^{ix}}{x} dx$$

as $R \rightarrow \infty$ and $\varepsilon \rightarrow 0$. Combining the two gives the result.

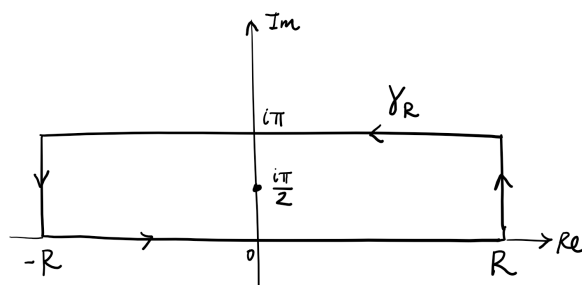


Example 13.4 – $\int_{\mathbb{R}} \frac{e^{ax}}{\cosh x} dx = \pi \sec(\pi a/2)$ where $-1 < a < 1$

Note that $\cosh(z) = 0$ iff $e^z + e^{-z} = 0$ iff $e^{2z} = -1$ iff $2z = (2n+1)i\pi$. We should avoid semicircular contours to avoid infinitely many poles in the limit. But $\cosh(x + i\pi) = -\cosh(x)$ so we can consider the following contour. Let $f(z) = e^{az}/\cosh(z)$. Along the right hand vertical segment,

$$\left| \int f(z) dz \right| = \left| \int_0^\pi \frac{e^{a(R+iy)}}{\cosh(R+iy)} i dy \right| \leq \int_0^\pi \frac{2e^{aR}}{|e^R - e^{-R}|} dy \rightarrow 0$$

as $R \rightarrow \infty$ since $a < 1$. On the left hand segment, we need $a > -1$ to control the integral. Combining all these gives the result.



Example 13.5 – $\int_0^\infty \frac{\log(x)}{1+x^2} dx = 0$

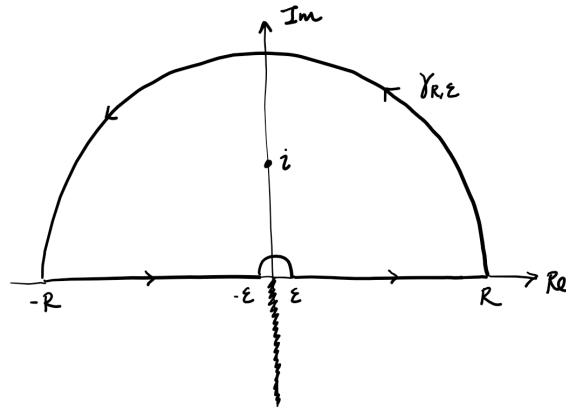
We need a cut plane to define $\log$. We can remove $i\mathbb{R}_{\leq 0}$ so that it does not interfere with our contour. On the large semicircle, the integrand is $\mathcal{O}((R \log R)/R^2) \rightarrow 0$ as $R \rightarrow \infty$. On the small semicircle, the integrand is $\mathcal{O}(\varepsilon \log \varepsilon) \rightarrow 0$ as $\varepsilon \rightarrow 0$. Along $\mathbb{R}_{< 0}$, $\log x = \log |x| + i\pi$. Cauchy's residue theorem for fixed $\gamma_{R,\varepsilon}$ says that $\int_{\gamma_{R,\varepsilon}} f(z) dz = 2\pi i \operatorname{res}_f(i)$ where $f(z) = \log(z)/(1+z^2)$. Assembling all of this, we

get

$$\int_0^\infty \frac{\log(x)}{1+x^2} dx + \int_{-\infty}^0 \frac{\log(x)}{1+x^2} (-dx) = 2\pi i \operatorname{res}_f(i)$$

$$2 \int_0^\infty \frac{\log(x)}{1+x^2} dx + i\pi \int_0^\infty \frac{1}{1+x^2} dx = \frac{i\pi^2}{2}$$

which gives the result. Note that splitting the integral into $\int_0^1$ and $\int_1^\infty$ then substituting gives a more direct way to do this.



A variation of the contour in Example 13.5 is the keyhole contour, where we remove $\mathbb{R}_{\geq 0}$ and use full circles instead of semicircles.

14 The argument principle

Zeros or poles of meromorphic functions have an order or multiplicity. Recall if $f : B(a, r)^* \rightarrow \mathbb{C}$, we can write $f(z) = (z - a)^k g(z)$ with $g : B(a, r) \rightarrow \mathbb{C}$ holomorphic and $g(a) \neq 0$.

Definition 14.1 – Order

If $k > 0$, we say that f has a zero of order k at a . If $k < 0$, we say that f has a pole of order $|k|$ at a . In both cases, we write the order as $\operatorname{ord}_f(a) > 0$.

Lemma 14.1

Suppose f is meromorphic on a domain U , with a zero (or respectively, a pole) of order k at $a \in U$. Then, $f'(z)/f(z)$ has a simple pole at a , of residue $\operatorname{ord}_f(a)$ (respectively, $-\operatorname{ord}_f(a)$).

Proof. If $f(z) = (z - a)^k g(z)$ and $g(a) \neq 0$, then $f'(z) = k(z - a)^{k-1} g(z) + (z - a)^k g'(z)$. So,

$$\frac{f'(z)}{f(z)} = \frac{k}{z - a} + \frac{g'(z)}{g(z)}$$

with gives the result since $g'(z)/g(z)$ is holomorphic near a . □

Definition 14.2 – Logarithmic derivative

The function $f'(z)/f(z)$ is called the logarithmic derivative of f .

If U is simply connected and $0 \notin U$, we can define a branch of $\log$ on U . Indeed, $1/z$ is holomorphic on U , so $\int_{\gamma} 1/z \, dz = 0$ for all closed γ by Cauchy's theorem for simply connected domains. By the converse to the fundamental theorem of calculus, there exists $L(z)$ such that $L(z)$ is holomorphic and is an antiderivative for $1/z$ on U . Pick $z_0 \in U$. We can set $e^{L(z_0)} = z_0$ by adding a constant if necessary. We check that $e^{L(z)}/z$ has derivative $e^{L(z)}(zL'(z) - 1)/z^2 = 0$. So, $e^{L(z)}/z$ is constant, and $e^{L(z)} = z$ on U . Hence, if f is non-vanishing and $\log f(z)$ makes sense on U , it has derivative f'/f .

Theorem 14.1 – The argument principle

Let U be a simply connected domain. Suppose f is meromorphic on U with zeroes $\{z_1, \dots, z_k\}$ and poles $\{w_1, \dots, w_l\}$. Let $\gamma : [a, b] \rightarrow U$ be a piecewise- C^1 closed path such that $z_i, w_j \notin \text{im}(\gamma)$ for all i, j . Then,

$$\frac{1}{2\pi i} \int_{\gamma} \frac{f'(z)}{f(z)} \, dz = \sum_{i=1}^k \text{ord}_f(z_i) I(\gamma, z_i) - \sum_{j=1}^l \text{ord}_f(w_j) I(\gamma, w_j).$$

Proof. The function f'/f has a singularity at $z \in U$ if and only if z is a zero or pole of f . Now combine Cauchy's residue theorem with Lemma 14.1. $\square$

Suppose γ bounds a domain D in U , so $I(\gamma, w) = 1$ if $w \in D$ and is zero otherwise. If γ is simple, then

$$\frac{1}{2\pi i} \int_{\gamma} \frac{f'}{f} \, dz = \text{number of zeroes of } f \text{ in } D - \text{number of poles of } f \text{ in } D$$

each counted according to their multiplicity. Let $\Gamma = f \circ \gamma$. In Theorem 14.1, the zeroes of f do not lie in γ , so Γ has image in $\mathbb{C}^*$. γ is closed so Γ is closed. Then,

$$\frac{1}{2\pi i} \int_{\gamma} \frac{f'(z)}{f(z)} \, dz = \frac{1}{2\pi i} \int_{\Gamma} \frac{1}{w} \, dw = I(\Gamma, 0).$$

Theorem 14.2 – Rouché's theorem

Let $U \subseteq \mathbb{C}$ be a domain. Let $\gamma \subseteq U$ be a closed curve which bounds a domain D in U . Let f and g be holomorphic on U , and suppose $|f(z)| > |g(z)|$ for all $z \in \gamma$. Then, f and $f + g$ have the same number of zeroes in D when counted with multiplicity.

Proof. $|f| > |g| \geq 0$ on γ , so neither f or $f + g$ can vanish on γ . Look at $h(z) = (f(z) + g(z))/f(z)$. Note that $h(z) \in B(1, 1)$ for all $z \in \gamma$. Thus, $h \circ \gamma$ has image in $\{\text{Re}(z) > 0\}$, so $I(h \circ \gamma, 0) = 0$. The argument principle says that the number of zeroes and the number of poles of h in D agree, counted with multiplicity. Since the zeroes of h corresponds to the zeroes of $f + g$, and the poles of h corresponds to the zeroes of f , this gives the result. $\square$

Example 14.1

$z^4 + 6z + 3$ has 3 roots in $\{1 < |z| < 2\}$ counted with multiplicity. On $|z| = 2$, $|z^4| = 16 > 6|z| + 3 \geq |6z + 3|$ so the polynomial has the same number of roots as z^4 in $\{|z| < 2\}$, which is 4. On $|z| = 1$, $6|z| = 6 > 4 \geq |z^2 + 3|$ so there exists only one root in $\{|z| < 1\}$. The results follows by the fundamental theorem of algebra.

The fundamental theorem of algebra tells us we have roots, and Rouché's theorem helps us locate them.

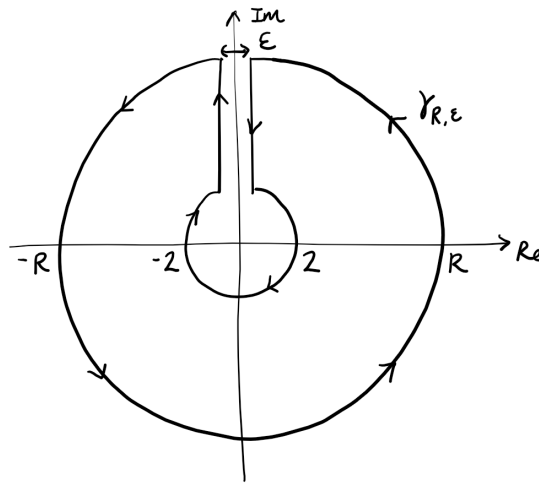
Example 14.2

Let $P(x) = x^n + a_{n-1}x^{n-1} + \dots + a_1x + a_0 \in \mathbb{Z}[x]$ with $a_0 \neq 0$ and $|a_{n-1}| > 1 + |a_{n-2}| + \dots + |a_1| + |a_0|$. We want to show that P is irreducible in $\mathbb{Z}[x]$ and hence in $\mathbb{Q}[x]$. Let $f(z) = a_{n-1}z^{n-1}$ and $g(z) = z^n +$

$a_{n-2}z^{n-2} + \cdots + a_1z + a_0$. Then, $|f| > |g|$ on $|z| = 1$, so f and $f + g = P$ have $n - 1$ roots in $\{|z| < 1\}$. So, P has only one root in $\{|z| \geq 1\}$. If $P(x) = Q(x)R(x)$ with $Q, R \in \mathbb{Z}[x]$ and $\deg(Q), \deg(R) \geq 1$, then at least one of Q or R has all roots in $\{|z| < 1\}$. Say all the roots of Q are in $\{|z| < 1\}$. But we assumed $a_0 \neq 0$, so 0 is not a root of P , and so is not a root of Q . But the product of the roots Q is a coefficient of Q , hence an integer strictly between 0 and 1. This is a contradiction.

Example 14.3 – $\int_{|z|=2} \frac{1}{z^5 + 12z - 5} dz = 0$

On $\gamma = \{|z| = 2\}$, $|z^5| = 32 > 29 \geq |12z - 5|$ so all the zeroes of $z^5 + 12z - 5$ live in $B(0, 2)$. Let $f(z) = 1/(z^5 + 12z - 5)$. Note that $\int_{|z|=R} f dz = \mathcal{O}(R^{-4}) \rightarrow 0$ as $R \rightarrow \infty$. The contour $\gamma_{R,\varepsilon}$ is homotopic to a constant loop in $\{|z| > 2 - \delta\}$, where we choose δ such that all poles lie in $\{|z| > 2 - \delta\}$. So, $\int_{\gamma_{R,\varepsilon}} f dz = 0$ by Cauchy. Now let $R \rightarrow \infty$ and $\varepsilon \rightarrow 0$.



15 Local degree

Definition 15.1 – Local degree

Let $a \in \mathbb{C}$ and $f : B(a, r) \rightarrow \mathbb{C}$ be holomorphic and non-constant. The local degree $\deg_{z=a}(f)$ of f at a is by definition the order of the zero at a of $g(z) := f(z) - f(a)$.

Lemma 15.1

Let $\gamma(t) = a + \varepsilon e^{it}$ with $t \in [0, 2\pi]$. If $\varepsilon > 0$ is sufficiently small, then $\deg_{z=a}(f) = I(f \circ \gamma, f(a))$.

Proof. By the identity theorem or the principle of isolated zeroes, since g is non-constant, there exists $\delta > 0$ such that g does not vanish on $B(a, \delta)^*$. Take $0 < \varepsilon < \delta$, then $I(f \circ \gamma, f(a))$ is well-defined since $f(a) \notin \text{im}(f \circ \gamma)$. Now apply the argument principle, using $I(g \circ \gamma, 0) = I(f \circ \gamma, f(a))$. $\square$

Theorem 15.1 – Local mapping degree

Let $f : B(a, r) \rightarrow \mathbb{C}$ be holomorphic and non-constant, and with $\deg_{z=a}(f) = k > 0$. Then, for all $\rho > 0$ sufficiently small, there exists $\varepsilon > 0$ such that if $0 < |w - f(a)| < \varepsilon$, then $f(z) = w$ has exactly k simple roots (local degree 1) in $B(a, \rho)$.

Proof. Pick $\rho > 0$ such that $g(z) = f(z) - f(a)$ and $g'(z) = f'(z)$ are both non-vanishing in $\overline{B(a, \rho)^*}$. This is possible by the principle of isolated zeroes. Let $\gamma(t) = a + \rho e^{2\pi i t}$ with $t \in [0, 1]$. Then, $f(a) \notin \text{im}(f \circ \gamma)$, so there exists $\varepsilon > 0$ such that $\text{im}(f \circ \gamma) \cap B(f(a), \varepsilon) = \emptyset$ since $f \circ \gamma$ has a closed and bounded image. If $w \in B(f(a), \varepsilon)$, we count solutions to $f(z) = w$ inside $B(a, \rho)$ using the argument principle. The number of zeroes of $f(z) - w$ in $B(a, \rho)$ is

$$\begin{aligned} I(f \circ \gamma, w) &= I(f \circ \gamma, f(a)) \quad \text{using that } I(f \circ \gamma, \cdot) \text{ depends only on the component of } \mathbb{C} \setminus \text{im}(f \circ \gamma) \text{ containing } \cdot \\ &= \deg_{z=a}(f) \\ &= k. \end{aligned}$$

So, we have k solutions with multiplicity, but we picked ρ such that f' is nonzero on $B(a, \rho)^*$, so all of these solutions must be simple. $\square$

Corollary 15.1 – Open mapping theorem

Let U be a domain and $f : U \rightarrow \mathbb{C}$ be holomorphic and non-constant. Then, f is an open map. That is, if $V \subseteq U$ is an open set, then $f(V)$ is open in $\mathbb{C}$.

Proof. We want to show that for all $z \in V$ and $\rho > 0$ sufficiently small (so $B(z, \rho) \subseteq V$), there exists $\varepsilon > 0$ such that $B(f(z), \varepsilon) \subseteq f(B(z, \rho)) \subseteq f(V) \subseteq \mathbb{C}$. The first inclusion is the local mapping degree theorem, noting that $k \geq 1$ in its statement. $\square$

Corollary 15.2

Let $U \subsetneq \mathbb{C}$ be a simply connected domain. Then, there exists a holomorphic injection from U to $\mathbb{D}$.

Proof. Let $q \in \mathbb{C} \setminus U$. Let $\phi(z) = z - q$, so $\phi : U \rightarrow \mathbb{C}$ is nowhere zero on U . Since U is simply connected, there is a branch of $\log(\phi)$ on U , so we can solve $\phi = h^2$ for some h holomorphic on U (there exists g such that $\phi = e^g$, then take $h = e^{g/2}$). Pick $y \in h(U)$. The open mapping theorem says that there exists $\varepsilon > 0$ such that $B(y, \varepsilon) \subseteq h(U)$. ϕ being injective implies that $h(U) \cap B(-y, \varepsilon) = \emptyset$ because if $h(z_1) = \pm h(z_2)$ then $z_1 = z_2$. Consider $z \mapsto \varepsilon / (2(h(z) + y))$ which is holomorphic on U , and the above shows that this has image in $\mathbb{D}$. $\square$

If $U = \mathbb{C}$, then Corollary 15.2 is false by Liouville's theorem.

16 Simply connected domains

Definition 16.1 – Local uniform convergence

Let $U \subseteq \mathbb{C}$ be open. Let $f_n : U \rightarrow \mathbb{C}$ be a sequence of holomorphic functions. We say (f_n) converges locally uniformly to f on U if for all $a \in U$, there exists $B(a, r) \subseteq U$ such that $f_n \rightarrow f$ uniformly on $B(a, r)$.

Example 16.1

$f_n(z) = z^n$ on $\mathbb{D}$ converge pointwise and locally uniformly to 0, but not uniformly. If $a \in \mathbb{D}$, and we pick r such that $|a| + r < 1$, then there exists $\rho < 1$ such that $B(a, r) \subseteq \overline{B(0, \rho)}$. Note that $|f_n(z)| < \varepsilon$ if and only if $|z| < \varepsilon^{1/n}$, so if we pick N sufficiently large such that $\rho < \varepsilon^{1/N} < 1$, then we see that (f_n) converges to 0 uniformly on $\overline{B(0, \rho)}$. However, it does not converge uniformly on $\mathbb{D}$.

Recall Corollary 7.2. In fact, we have the following slightly stronger statement.

Corollary 16.1

Let $f_n : U \rightarrow \mathbb{C}$ be a sequence of holomorphic functions on a domain U , and suppose $f_n \rightarrow f$ uniformly on U . Then, f is holomorphic and $f'_n \rightarrow f'$ locally uniformly.

Proof. Recall that for $0 < \rho < r$ and $w \in B(a, \rho)$, we have

$$|f'_n(w) - f'(w)| = \frac{1}{2\pi} \left| \int_{|z-a|=\rho} \frac{f_n(z) - f(z)}{(z-w)^2} dz \right|.$$

If $|w - a| \leq \rho/2$, then

$$|f'_n(w) - f'(w)| \leq \frac{1}{2\pi} (2\pi\rho) \sup_{|z-a|=\rho} |f_n(z) - f(z)| \rightarrow 0$$

by uniform convergence of $f_n \rightarrow f$. This estimate is uniform in w so $f'_n \rightarrow f'$ locally uniformly. $\square$

Proposition 16.1

Let $f_n : U \rightarrow \mathbb{C}$ be a sequence of holomorphic functions on a domain U , and suppose $f_n \rightarrow f$ uniformly on U . If f_n is injective for all n , then f is injective or constant.

Proof. Suppose for contradiction that f is non-constant, but there exists $z_1 \neq z_2 \in U$ such that $f(z_1) = f(z_2) = a \in \mathbb{C}$. Let z_1 and z_2 be joined by a polygonal path in U as in the proof of Proposition 2.1. Construct a closed curve γ in U , bounding a domain in U containing z_1 and z_2 . Without loss of generality, assume $f(z) \neq a$ for all $z \in \gamma$, since $f^{-1}(a)$ consists of isolated points of U by the identity theorem. The argument principle gives

$$1 \leq \frac{1}{2\pi i} \int_{\gamma} \frac{f'_n(z)}{f_n(z) - a} dz$$

since f_n is injective for all n . So,

$$1 \leq \frac{1}{2\pi i} \int_{\gamma} \frac{f'(z)}{f(z) - a} dz$$

since $f_n \rightarrow f$ and $f'_n \rightarrow f'$ locally uniformly, and hence on the closed bounded set bound by γ . But by the argument principle,

$$\frac{1}{2\pi i} \int_{\gamma} \frac{f'(z)}{f(z) - a} dz \geq 2$$

since $f(z_1) = f(z_2) = a$. This is a contradiction. $\square$

Theorem 16.1 – Montel's theorem

Let $\mathcal{F}$ be a family of holomorphic functions on a domain U . Assume $\mathcal{F}$ is uniformly bounded, that is, there exists $R > 0$ such that $\text{im}(f) \subseteq B(0, R)$ for all $f \in \mathcal{F}$. If $(f_j) \subseteq \mathcal{F}$ is any sequence from $\mathcal{F}$, then it has a locally uniformly convergent subsequence.

A fundamental theorem in real analysis says that if $(f_n : [a, b] \rightarrow \mathbb{R})$ are uniformly bounded and equicontinuous (that is, for all $\varepsilon > 0$, there exists $\delta > 0$ such that for all n , $|x - y| < \delta$ implies $|f_n(x) - f_n(y)| < \varepsilon$), then there exists a subsequence converging uniformly.

Theorem 16.2 – Riemann mapping theorem

If $U \subsetneq \mathbb{C}$ is a simply connected domain, then there exists a conformal equivalence from U to $\mathbb{D}$.

Proof. Fix $z_0 \in U$. Let $\mathcal{F} = \{f : U \rightarrow \mathbb{D} \mid \text{holomorphic, injective, and } z_0 \mapsto 0\}$. The proof has three steps.

(i) $\mathcal{F} \neq \emptyset$.

(ii) There exists $f_\infty \in \mathcal{F}$ such that $|f'_\infty(z_0)|$ is the maximal value this achieves across all $f \in \mathcal{F}$.

(iii) f_∞ is surjective, and defines the required conformal isomorphism.

For (i), we use Corollary 15.2. If $f(z_0) \neq 0$, we can compose it with a Mobius map to send $f(z_0)$ to 0.

For (ii), pick $(f_n) \subseteq \mathcal{F}$ such that $|f'_n(z_0)| \rightarrow \sup_{f \in \mathcal{F}} |f'(z_0)|$. $\mathcal{F}$ is a uniformly bounded family, so there exists a subsequence $f_{n_j} \rightarrow f_\infty$ locally uniformly by Montel's theorem. Then, $|f'_\infty(z_0)| = \max_{f \in \mathcal{F}} |f'(z_0)| > 0$, so f_∞ is not constant, so it is injective and $f_\infty \in \mathcal{F}$ by Proposition 16.1.

For (iii), if f_∞ is not onto $\mathbb{D}$, say $\alpha \in \mathbb{D} \setminus f_\infty(U)$. Consider $\sigma_\alpha(z) = (z - \alpha)/(\bar{\alpha}z - 1)$ which is a Mobius map that maps $\mathbb{D}$ to itself. $(\sigma_\alpha \circ f_\infty)(U)$ is a simply connected domain in $\mathbb{D}$ not containing 0. So, there exists a holomorphic branch of $\log(\sigma_\alpha \circ f_\infty)$ on U , so we can let $s(z)$ be the square root of $\sigma_\alpha \circ f_\infty$. Let $h(z) = \sigma_{s(\alpha)} \circ s \circ (\sigma_\alpha \circ f_\infty) : U \rightarrow \mathbb{D}$. Since $s^2 = \sigma_\alpha \circ f_\infty$, we can check that $h \in \mathcal{F}$. Write $f_\infty = \Phi \circ h$ where $\Phi = \sigma_\alpha^{-1} \circ (\text{squaring}) \circ \sigma_{s(\alpha)}^{-1}$. Now $\Phi : \mathbb{D} \rightarrow \mathbb{D}$ is holomorphic, maps $0 \mapsto 0$, and is not a rotation. By Schwarz's lemma, $|\Phi'(0)| < 1$. Thus, $|f'_\infty(z_0)| < |h'(z_0)|$ which is a contradiction. Hence, f_∞ is surjective, and $U \cong \mathbb{D}$. $\square$

What do these maps look like? In fact, these maps are not expressible using elementary functions. Is it useful? In algebraic topology and Riemann surfaces, many interesting holomorphic objects are $\mathbb{D}/G$, where G is a group acting holomorphically on $\mathbb{D}$.